

DAYSTAR UNIVERSITY
SCHOOL OF SCIENCE, ENGINEERING AND HEALTH
DEPARTMENT OF SCIENCE AND ENGINEERING

**COMPARING LEE-CARTER AND LONG SHORT-TERM MEMORY
MODELS FOR AGE-SPECIFIC MORTALITY FORECASTING IN
KENYA: 2025-2030 PROJECTIONS**

**A RESEARCH PROJECT SUBMITTED IN PARTIAL FULFILMENT OF
THE REQUIREMENTS FOR THE AWARD OF THE DEGREE OF
BACHELOR OF SCIENCE IN ACTUARIAL SCIENCE**

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DECLARATION

I, Zuber Haji (ADM 22-2333), declare that this research project, “Comparing Lee-Carter and Long Short-Term Memory Models For Age-Specific Mortality Forecasting in Kenya: 2025-2030 Projections,” is my **own original work and has not been submitted, in whole or in part, for the award of a degree or** any other academic qualification at Daystar University or at any other institution of higher learning. All sources of information used or quoted in this project have been duly acknowledged by means of complete references, in accordance with the requirements of Daystar University's Department of Actuarial Science.

DEDICATION

I lovingly dedicate this research project to the cherished memories of my late father, **Said Haji**, and my late grandfather, **Omar Buya**.

Although you were not here to witness the completion of this journey, every step I have taken and every hardship I have overcome has been shaped by the strength, encouragement, and guidance you instilled in me. You showed me the light when I needed it most, and your love and values continue to guide me even in your absence. I once read that writing someone's name allows them to live forever. In my own small way, that is what I hope to do here. May your names continue to be remembered, your legacy continue to inspire, and may the beauty of the afterlife continue to surround you both.

You are deeply missed, forever loved, and never forgotten.

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Any errors or shortcomings that remain in this work are entirely my own.

Table of Contents

DECLARATION	i
DEDICATION	ii
ACKNOWLEDGEMENT	iii
LIST OF TABLES	vii
LIST OF FIGURES	viii
LIST OF ACRONYMS AND ABBREVIATIONS	ix
ABSTRACT	x
CHAPTER ONE	1
INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	2
1.3 Objectives of the Study	3
1.3.1 General Objective	3
1.3.2 Specific Objectives	4
1.4 Research Questions	4
1.5 Significance of the Study	4
1.6 Scope of the Study	5
1.7 Limitations of the Study	6
CHAPTER TWO	8
LITERATURE REVIEW	8
2.1 Introduction	8
2.2 Conceptual Background	8
2.3 Theoretical Framework	9
2.3.1 The Lee-Carter Model	9
2.3.2 The LSTM Architecture	10
2.3.3 Theoretical Rationale	10
2.3.4 Comparability of Statistical and Data-Science Forecasts: Justifying a Common Accuracy Metric	11
2.4 Critical Analysis of Literature Review	13

2.5 Research Gap Identification.....	14
2.6 Development of Research Framework	15
2.7 Variable Definitions.....	16
2.8 Hypothesis Development.....	17
2.9 Conceptual Model.....	17
CHAPTER THREE	19
RESEARCH METHODOLOGY	19
3.1 Introduction	19
3.2 Data and Data Sources.....	19
3.3 Data Preparation.....	21
3.4 Model I: The Lee-Carter Stochastic Mortality Model	22
3.4.1 Model Specification	22
3.4.2 Parameter Estimation	22
3.4.3 Forecasting the Mortality Index.....	23
3.5 Model II: The LSTM Neural Network Model	23
3.5.1 Model Rationale and Input Structure	23
3.5.2 Cell Architecture	24
3.5.3 Network Training	26
3.6 Model Implementation	27
3.7 Model Validation and Out-of-Sample Testing	27
3.8 Forecast Accuracy Metric.....	28
3.8.1 Root Mean Square Error (RMSE)	28
3.8.2 Mean Absolute Error (MAE).....	29
3.8.3 Mean Absolute Percentage Error (MAPE)	29
3.9 Statistical Test for Model Comparison.....	30
3.10 Ethical and Data Considerations	31
CHAPTER FOUR	32
DATA ANALYSIS AND FINDINGS	32
4.1 Introduction	32

4.2 Historical Mortality Trends in Kenya, 1990-2024	32
4.3 Age-Specific Mortality Forecasts from the LSTM Model, 2025-2030.....	33
4.4 Age-Specific Mortality Forecasts from the Lee-Carter Model, 2025-2030	36
4.5 Comparative Forecast Accuracy: LSTM versus Lee-Carter	40
4.6 Statistical Significance of the Accuracy Difference: The Diebold-Mariano Test.....	42
CHAPTER FIVE	45
DISCUSSION, CONCLUSION AND RECOMMENDATIONS.....	45
5.1 Introduction	45
5.2 Discussion.....	45
5.2.1 LSTM Mortality Forecasting	45
5.2.2 Lee-Carter Mortality Forecasting.....	46
5.2.3 Comparative Model Performance	47
5.2.4 Age-Specific Mortality Projections.....	47
5.3 Conclusions	48
5.3.1 LSTM Mortality Forecasting	48
5.3.2 Lee-Carter Mortality Forecasting.....	49
5.3.3 Comparative Model Performance	49
5.3.4 Age-Specific Mortality Projections.....	49
5.4 Recommendation.....	49
5.5 Limitations of the Study	50
REFERENCES	51
APPENDICES	53
Appendix A: Supporting Datasets	53
Appendix B: Lee-Carter Estimated Age Parameters	54
Appendix C: Lee-Carter Forecasted Age-Specific Mortality Rates, 2025-2030	54
Appendix D: LSTM Forecasted Age-Specific Mortality Rates, 2025-2030	55
Appendix E: Analysis Code	56

LIST OF TABLES

Table 2. 1: What Varies Across the Two Models, and What Stays Constant for Comparison	12
Table 3. 1: Summary of Data Sources	20
Table 4. 1: LSTM Forecasted Age-Specific Mortality Rates for Selected Ages, Kenya, 2025-2030.....	35
Table 4. 2: Forecasted Mortality Index (k_t), 2025-2030	39
Table 4. 3: Lee-Carter Forecasted Age-Specific Mortality Rates for Selected Ages, Kenya, 2025-2030.....	39
Table 4. 4: Forecast Accuracy Comparison, LSTM vs Lee-Carter (2023-2024 Hold-Out)	41
Table 4. 5: Model Comparison Summary	41
Table 4. 6: Diebold-Mariano Test of Equal Predictive Accuracy	43

LIST OF FIGURES

Figure 3. 1: Architecture of a Long Short-Term Memory (LSTM) Cell	26
Figure 4. 1: Log-Mortality Heatmap for Kenya by Age and Year, 1990-2024	33
Figure 4. 2: LSTM Training History	34
Figure 4. 3: LSTM-Forecasted Age-Specific Mortality Rates by Age, Kenya, 2025-2030	35
Figure 4. 4: Estimated Average Log-Mortality Parameter (a_x , top) and Age-Sensitivity Parameter (b_x , bottom) by Age	37
Figure 4. 5: Estimated Mortality Index (k_t) for Kenya, 1990-2024	38
Figure 4. 6: Forecast of the Mortality Index (k_t) for Kenya, 2025-2030	38
Figure 4. 7: Lee-Carter Forecasted Age-Specific Mortality Rates by Age, Kenya, 2025-2030 ...	40
Figure 4. 8: Lee-Carter's Accuracy Advantage Over LSTM, by Metric (2023-2024 Hold-Out)	42
Figure 4. 9: Diebold-Mariano Test - Observed Statistic Against the Standard Normal Distribution	43

LIST OF ACRONYMS AND ABBREVIATIONS

Acronym	Meaning
ARIMA	Autoregressive Integrated Moving Average
ASMR	Age-Specific Mortality Rate
DM	Diebold-Mariano (test)
GHE	Global Health Estimates (WHO)
HIV/AIDS	Human Immunodeficiency Virus / Acquired Immunodeficiency Syndrome
IRA	Insurance Regulatory Authority
KDHS	Kenya Demographic and Health Survey
KNBS	Kenya National Bureau of Statistics
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
MSE	Mean Squared Error
NSSF	National Social Security Fund
RMSE	Root Mean Square Error
RNN	Recurrent Neural Network
SVD	Singular Value Decomposition
TB	Tuberculosis
UN	United Nations
WHO	World Health Organization
WPP	World Population Prospects (UN)

ABSTRACT

Accurate forecasting of age-specific mortality rates is essential for actuarial pricing, pension planning, and public health resource allocation. Kenyan mortality forecasting has relied predominantly on the Lee-Carter model, which assumes a linear time trend and may struggle to capture the non-linear mortality dynamics of Kenya's demographic transition. Long Short-Term Memory (LSTM) neural networks have emerged globally as an alternative capable of learning non-linear patterns directly from data, matching or outperforming classical models in most countries studied. However, Kenyan evidence is limited and mixed: the only identified Kenyan comparison of a deep-learning model against a classical benchmark, applied to a health-burden indicator rather than mortality, found the classical model performed better. No study had benchmarked LSTM against Lee-Carter using Kenya's own age-specific mortality data.

This study addressed that gap by applying an LSTM model to forecast Kenya's age-specific mortality rates for 2025-2030, using single-year mortality data (ages 0-100) for 1990-2024 from the UN World Population Prospects 2024 database, benchmarked against Lee-Carter. Both models were trained on 1990-2019, validated on 2020-2022, and evaluated on a 2023-2024 hold-out sample using RMSE, MAE, MAPE, and the Diebold-Mariano test.

Lee-Carter significantly outperformed LSTM on all three metrics (RMSE: 0.0091 versus 0.1084; MAE: 0.0059 versus 0.0744; MAPE: 27.83% versus 2,276.46%), confirmed by the Diebold-Mariano test ($DM = -7.294$, $p\text{-value} < 0.001$). This diverges from the global literature but aligns with the only other Kenyan deep-learning-versus-classical comparison (Karani & Dongxiao, 2025), suggesting LSTM's accuracy advantage does not necessarily transfer to Kenya's shorter, structurally disrupted mortality record. Lee-Carter's forecasts were adopted as the study's final projections for 2025-2030, indicating a gradual rise, rather than continued decline, in mortality across the age range. These findings offer locally grounded evidence for actuaries, insurers, and policymakers on the relative suitability of classical and machine-learning mortality forecasting approaches in Kenya.

CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Mortality forecasting is a cornerstone of demographic planning, actuarial pricing, and the allocation of public health resources. Accurate projections of age-specific mortality rates (ASMRs) enable governments, insurers, and pension schemes to anticipate future population health burdens, price life-contingent products appropriately, and design sustainable social protection systems. Globally, mortality modelling has evolved from simple extrapolative techniques to sophisticated stochastic and, more recently, machine-learning-based approaches capable of capturing non-linear temporal dependencies in mortality trends.

Kenya has undergone a marked demographic and epidemiological transition over the past three decades. Life expectancy at birth has risen steadily, with recent estimates placing it at roughly 64-68 years depending on the data source and methodology used (World Bank, 2024; Central Intelligence Agency, 2024), up from considerably lower levels in earlier decades. Crude death rates have similarly declined to approximately 5.0 per 1,000 population in recent years (Central Intelligence Agency, 2024). However, this aggregate improvement conceals substantial heterogeneity across age groups: under-five mortality, while declining, remains a significant contributor to overall mortality burden, while non-communicable diseases have become an increasingly dominant cause of death among older adults, alongside a continuing burden from communicable diseases such as HIV/AIDS, tuberculosis, and malaria. Kenya's population also remains youthful, with a median age of under 20 years (United Nations, 2024), meaning the age structure driving future mortality experience will shift considerably over the coming decades.

Traditional mortality models, such as the Lee-Carter model and its extensions, have been applied in the Kenyan and broader African contexts to model and forecast national mortality trends. While these stochastic models offer interpretability and a strong theoretical foundation grounded in demographic and actuarial science, they generally assume linear or log-linear time trends in mortality indices and can struggle to capture complex, non-linear, and abrupt shifts in mortality

patterns, such as those induced by disease outbreaks, health policy interventions, or socioeconomic shocks.

Deep learning methods, and Long Short-Term Memory (LSTM) networks in particular, have emerged as a promising alternative for time-series forecasting problems characterized by long-range temporal dependencies and non-linear dynamics. LSTM networks, a class of recurrent neural networks equipped with memory cells and gating mechanisms, are designed to retain relevant historical information over extended sequences while mitigating the vanishing gradient problem that limits standard recurrent networks. These properties make LSTM models well-suited to mortality data, which exhibit both long-term structural trends and shorter-term fluctuations across age cohorts. Studies in other national contexts have demonstrated that LSTM and related deep learning architectures can match or outperform classical stochastic mortality models in forecast accuracy, particularly over longer horizons.

Despite this global momentum, empirical application of deep learning techniques to age-specific mortality forecasting in Kenya, and in Sub-Saharan Africa more broadly, remains limited. Much of the existing Kenyan mortality literature relies on classical demographic and stochastic approaches, and data quality constraints-incomplete vital registration coverage and reliance on modelled WHO or UN estimates-further complicate mortality analysis. This study therefore seeks to apply LSTM deep learning models to historical age-specific mortality data in Kenya in order to generate five-year mortality projections for the period 2025-2030, contributing both a methodological extension and a practically useful forecasting tool to the Kenyan demographic and actuarial literature.

1.2 Statement of the Problem

Reliable age-specific mortality projections are essential for actuarial pricing of life insurance and annuity products, the design of pension and social security schemes, national healthcare planning, and efficient resource allocation. In Kenya, however, mortality forecasting has traditionally relied on classical stochastic models and, in some cases, on mortality tables adapted from other jurisdictions or developed using relatively limited historical mortality data.

While these approaches have been widely applied in actuarial and demographic practice, recent advances in machine learning have introduced alternative forecasting methods capable of modelling complex nonlinear temporal relationships. Nevertheless, evidence regarding the suitability of these approaches for Kenya's mortality experience remains limited.

This creates an important knowledge gap. Although numerous international studies have reported improved forecasting performance using deep learning models, particularly Long Short-Term Memory (LSTM) networks, these findings have been obtained predominantly from countries with long, complete, and high-quality historical mortality records. Their applicability to Kenya, where the historical mortality series is comparatively shorter and reflects different demographic and epidemiological conditions, has not been established. Consequently, actuaries, insurers, pension managers, and policymakers lack empirical evidence on whether modern deep learning models provide more accurate mortality forecasts than established stochastic approaches when applied to Kenya's historical age-specific mortality data.

Without such evidence, the selection of mortality forecasting models for actuarial valuation, insurance pricing, pension reserving, and public health planning may not be based on the most appropriate methodology for the Kenyan context. This study addresses this gap by comparing the forecasting performance of the Lee-Carter stochastic mortality model and a Long Short-Term Memory (LSTM) neural network using Kenya's historical age-specific mortality data. The study also generates age-specific mortality projections for Kenya for the period 2025-2030 based on the forecasting model that demonstrates superior predictive performance.

1.3 Objectives of the Study

1.3.1 General Objective

To apply a Long Short-Term Memory (LSTM) deep learning model, benchmarked against the Lee-Carter stochastic mortality model, in forecasting age-specific mortality rates in Kenya for the period 2025-2030, using historical mortality data covering the period 1990-2024.

1.3.2 Specific Objectives

1. To apply an LSTM model in forecasting age-specific mortality rates in Kenya for the period 2025-2030, using historical age-specific mortality data covering the period 1990-2024.
2. To compare the forecasting accuracy of the LSTM model against the Lee-Carter stochastic mortality model, using standard error metrics (RMSE, MAE, and MAPE), based on historical mortality data covering the period 1990-2024.

1.4 Research Questions

1. How effectively can an LSTM model be applied to forecast age-specific mortality rates in Kenya using historical mortality data across different age groups?
2. How does the forecasting accuracy of the LSTM model compare with that of a traditional stochastic mortality model (such as the Lee-Carter model) when applied to Kenyan mortality data?
3. What are the projected age-specific mortality rates for Kenya over the five-year period from 2025 to 2030, and what trends do these projections reveal across different age groups?

1.5 Significance of the Study

To the Actuarial Profession: The study offers actuaries in Kenya and the broader East African region an empirically tested, data-driven alternative to conventional mortality models for pricing life insurance, annuities, and other mortality-contingent products, as well as for setting reserves and conducting experience investigations. A locally derived, deep-learning-based mortality projection tool can improve the precision of actuarial assumptions and reduce reliance on foreign mortality tables that may not reflect Kenya's specific mortality experience.

To Policy Makers: Accurate age-specific mortality projections support evidence-based national planning, including social security and pension scheme design, public health budgeting, and population policy formulation. The findings can inform institutions such as the National Treasury, the National Social Security Fund (NSSF), the Insurance Regulatory Authority (IRA), and the Ministry of Health in anticipating future demographic and health burdens.

To the Healthcare Sector: Healthcare planners and public health practitioners can use the projected age-specific mortality rates to identify vulnerable population groups, anticipate future healthcare needs, and improve the allocation of medical resources. The findings may also support the evaluation of existing health interventions and the formulation of strategies aimed at reducing preventable mortality.

To Research and Academia: The study contributes to the relatively sparse body of literature applying deep learning methods to mortality forecasting in Sub-Saharan Africa, providing a methodological reference and empirical benchmark for future researchers.

It also opens avenues for further research combining machine learning with demographic and actuarial science in data-constrained developing-country contexts.

Daystar University and the Academic Community: The study enriches the existing body of knowledge within the fields of actuarial science, statistics, and data science. It also demonstrates the practical application of deep learning techniques in solving real-world demographic problems, thereby supporting teaching, learning, and future academic research.

1.6 Scope of the Study

This study focuses on forecasting age-specific mortality rates in Kenya at the national level using historical age-specific mortality data obtained from the United Nations World Population Prospects (WPP) 2024 database for the period 1990-2024. The study compares the forecasting performance of the Lee-Carter stochastic mortality model and a Long Short-Term Memory (LSTM) neural network using the same historical mortality dataset. Forecast accuracy is evaluated using RMSE, MAE, MAPE, and the Diebold-Mariano test, after which age-specific mortality projections are generated for the period 2025-2030.

The scope of the study is limited to national-level mortality forecasting and does not include sub-national (county-level) mortality analysis, cause-specific mortality modelling, or the pricing and valuation of actuarial products. Nevertheless, the projected mortality rates generated by the study may provide valuable input for future actuarial, demographic, and public health applications.

1.7 Limitations of the Study

1. Data quality and completeness: Kenya's vital registration system has historically had incomplete coverage, particularly for deaths occurring outside formal health facilities and in rural areas. Consequently, the study may rely partly on modelled estimates (e.g., from KNBS, UN World Population Prospects, or WHO), which carry their own estimation uncertainty and may not perfectly reflect actual mortality experience.
2. Length of historical time series: Deep learning models such as LSTMs typically perform best with long historical sequences. Kenya's reliable time-series mortality data may be relatively short, which can constrain model training and increase the risk of overfitting, requiring careful regularization and validation strategies.
3. Forecast horizon: The study projects mortality rates only for the period 2025-2030. Although this short-term forecasting horizon is appropriate for evaluating model performance and supporting near-term actuarial and policy planning, the results may not necessarily reflect longer-term mortality dynamics beyond the study period.
4. Model interpretability: LSTM models are inherently less interpretable ("black-box") than classical stochastic mortality models, which may limit their acceptance among practitioners who require transparent, explainable assumptions for regulatory or pricing purposes.
5. Financial constraints: The study was conducted within limited financial resources, which constrained access to certain proprietary mortality datasets, specialised software, and subscription-based academic publications that could have provided additional data or methodological insights. Consequently, the study relied primarily on publicly available and institutional data sources considered appropriate for achieving the research objectives.
6. Time constraints: The study was undertaken within the time frame prescribed for the undergraduate research project. This limited the opportunity to explore a wider range of forecasting models, conduct extensive sensitivity analyses, or evaluate longer forecast horizons. Nevertheless, careful planning and adherence to the research schedule ensured that the study objectives were achieved within the available time.

7. Exclusion of exogenous shocks: The forecasting model is based on historical mortality patterns and may not fully anticipate the impact of unforeseen future shocks (e.g., pandemics, major policy shifts, or economic crises) on mortality rates during the 2025-2030 projection period.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Reliable forecasts of age-specific mortality rates underpin actuarial pricing, pension and social security planning, and public health resource allocation. Mortality forecasting has traditionally relied on stochastic demographic models, most notably the Lee-Carter model, which assumes mortality change follows a single linear time trend. In recent years, deep learning methods - particularly Long Short-Term Memory (LSTM) neural networks have emerged as an alternative capable of learning non-linear temporal patterns directly from historical mortality data, without imposing this linear assumption. This chapter reviews the conceptual, theoretical, and empirical literature relevant to comparing these two approaches for forecasting Kenya's age-specific mortality rates. It proceeds through the conceptual background underpinning mortality forecasting, the theoretical frameworks the study draws on, a critical analysis of recent empirical literature, the specific research gap that analysis reveals, the research framework developed from it, formal definitions of the study's variables, the study's hypotheses, a conceptual model, and a chapter summary. Particular attention is paid throughout to how existing historical mortality data, as distinct from newly collected data, has been used and evaluated in the literature, and citations are drawn predominantly from the last five years (2021-2026), with the three foundational methodological papers underlying the study's models and evaluation framework retained as necessary exceptions (Lee & Carter, 1992; Hochreiter & Schmidhuber, 1997; Diebold & Mariano, 1995).

2.2 Conceptual Background

Mortality forecasting rests on a small set of fundamental concepts that recur throughout this study. An **age-specific mortality rate (ASMR)** is defined as the number of deaths at age x in year t divided by the mid-year population exposed to risk at that age and year. It represents the fundamental unit of analysis and forecasting in this study and is distinct from the aggregate crude death rate.

Historical mortality data refers to records of age-specific mortality rates that have already been observed and compiled by official statistical agencies over previous years. The distinction between utilising existing mortality records and generating new mortality data through primary data collection is important in understanding the sources and nature of the data employed in mortality forecasting.

A **mortality forecast** is the projection of age-specific mortality rates into future time periods using a model calibrated on historical mortality data. The **Lee-Carter model** and the **Long Short-Term Memory (LSTM) network** constitute the two principal modelling approaches considered in this study, representing traditional stochastic and modern deep learning approaches to mortality forecasting, respectively.

Finally, **forecast accuracy** is evaluated using the **Root Mean Square Error (RMSE)**, **Mean Absolute Error (MAE)**, and **Mean Absolute Percentage Error (MAPE)**. These performance measures provide quantitative assessments of forecasting errors and are widely used in empirical studies to evaluate and compare the predictive performance of mortality forecasting models.

2.3 Theoretical Framework

2.3.1 The Lee-Carter Model

The Lee-Carter model, introduced by Lee and Carter (1992), decomposes the logarithm of ASMRs into an age-specific average pattern, an age-specific sensitivity parameter, and a single time-varying mortality index, conventionally forecast forward using a random walk with drift. A recent bibliometric mapping of the field confirms the model remains the most heavily cited stochastic mortality framework, even as machine-learning alternatives grow (Redzwan & Ramli, 2022). Its central theoretical weakness, assuming mortality change follows a single linear trend across all age groups, is what motivates comparison against a more flexible, data-driven alternative.

The Lee-Carter model is commonly expressed as

$$\log m(x, t) = a_x + b_x k_t + \varepsilon_{x,t}$$

Where:

- $m(x,t)$ = is the mortality rate for age x in year t ,
- a_x = represents the average mortality patterns across age groups,
- b_x = is the age-specific sensitivity to changes in the mortality index,
- k_t = is the overall mortality trend over time,
- $\epsilon_{x,t}$ = represents the error term.

The Lee-Carter model remains highly relevant to the present study because it provides a well-established theoretical and methodological benchmark for mortality forecasting. Its widespread acceptance in demographic and actuarial research makes it an appropriate reference model for evaluating the forecasting performance of more flexible approaches, including deep learning models such as Long Short-Term Memory (LSTM) networks.

2.3.2 The LSTM Architecture

The Long Short-Term Memory (LSTM) architecture, introduced by Hochreiter and Schmidhuber (1997), extends recurrent neural networks with a gated memory cell that retains relevant information across long sequences while discarding irrelevant information, overcoming the vanishing-gradient limitation of standard recurrent networks. This gating mechanism allows an LSTM to hold long-run trend information while remaining sensitive to shorter-run deviations, a dual structure well matched to mortality data, which combines decades-long secular decline with shorter, sometimes abrupt, deviations, such as the disruption to Kenya's mortality decline documented around the 1990s HIV/AIDS epidemic (Njenga & Kipchirchir, 2024). This property has motivated a growing body of work applying LSTM directly to mortality time series rather than only to derived indices (Perla et al., 2021).

2.3.3 Theoretical Rationale

Together, these two frameworks provide competing theoretical accounts of how a historical mortality record should be used to generate a forecast: Lee-Carter assumes the record's future continuation follows a single linear index, whereas LSTM assumes the record's temporal structure can be learned directly, without that assumption.

LSTM is treated as the primary analytical lens in this review, with Lee-Carter retained as the benchmark inherited from the dominant tradition in the field, a positioning consistent with recent work that combines both frameworks directly, such as embedding an LSTM within the Lee-Carter mortality-index forecast (Marino et al., 2022), though the present study tests the two independently rather than nesting one within the other.

2.3.4 Comparability of Statistical and Data-Science Forecasts: Justifying a Common Accuracy Metric

Sections 2.3.1 and 2.3.2 establish that the Lee-Carter and LSTM models belong to two different methodological paradigms: the former a classical statistical model, estimated in closed form and governed by explicit distributional assumptions; the latter a data-science method, estimated iteratively through gradient-based optimisation with no prior parametric structure imposed. This raises a methodological question that must be addressed before the two models can be meaningfully compared: whether it is appropriate to evaluate a statistical model and a data-science model using the same accuracy metrics, given that they are estimated in fundamentally different ways.

The answer, drawn from both the logic of forecast evaluation and established practice in the comparative mortality-forecasting literature, is yes. RMSE, MAE, and MAPE, and the Diebold-Mariano test built upon them, do not evaluate how a forecast was produced; they evaluate how close the forecast came to what actually happened. Each metric is computed purely from paired observations, the forecasted mortality rate and the corresponding observed mortality rate, over the same hold-out period, on the same scale. Nothing in their construction refers to whether the underlying model was estimated via Singular Value Decomposition or backpropagation through time. This is precisely what makes them suitable for cross-paradigm comparison: a common accuracy metric requires only that both models generate a forecast for the same target on the same data, not that they share an estimation philosophy. Indeed, no comparison between a statistical and a data-science model would be possible at all without such a paradigm-neutral yardstick, since there is no meaningful sense in which the “goodness” of a matrix decomposition can be measured against the “goodness” of a trained neural network except by asking which one predicted reality more accurately.

Table 2. 1: What Varies Across the Two Models, and What Stays Constant for Comparison

Held constant across both models	Allowed to vary between models
Historical mortality panel (1990-2024)	Estimation method (SVD vs gradient-based optimisation)
Chronological train/validation/test partition	Internal model structure (parametric index vs gated cells)
Hold-out evaluation period (2023-2024)	Assumptions about the age-time relationship
Accuracy metrics applied (RMSE, MAE, MAPE, DM test)	Interpretability and estimation transparency

This is also the convention followed throughout the comparative literature this study draws on. Chen and Khaliq (2023) benchmarked LSTM, Bidirectional LSTM, and Gated Recurrent Unit models directly against Lee-Carter using standard error metrics; Marino et al. (2022) and Perla et al. (2024) evaluated neural and hybrid architectures against classical stochastic benchmarks on the same basis; and, most directly relevant to this study, Karani and Dongxiao (2025) compared ARIMA, a classical statistical model, against LSTM, a data-science model, using RMSE, MAE, MAPE, and the Diebold-Mariano test, the identical evaluation framework adopted here. None of these studies treated the difference in estimation paradigm as a barrier to comparison; all treated predictive accuracy on held-out data as the relevant, paradigm-neutral basis for it. The Diebold-Mariano test itself was developed on exactly this premise (Diebold & Mariano, 1995): it operates on the loss differential between two competing forecasts, drawing no distinction between how either forecast was generated.

This does not mean the comparison is free of methodological caveats, only that the caveats concern the fairness of the experimental setup rather than the validity of the metrics themselves. As Section 2.4 discusses, LSTM models typically require longer historical series than classical models to show their full accuracy advantage (Lindholm & Palmborg, 2022), and Kenya’s 35-year series is short by the standards of the century-scale series used in most published LSTM mortality studies. If LSTM underperforms in this study, that may reflect this data constraint rather than any general inferiority of data-science approaches to mortality forecasting.

The metrics themselves, however, remain the correct tool for detecting whatever difference is present: they measure predictive accuracy against observed outcomes, which is the only basis on which a statistical model and a data-science model can be compared on equal terms.

2.4 Critical Analysis of Literature Review

The empirical literature relevant to this review can be read along two complementary lines: what has been found about LSTM's forecasting performance relative to Lee-Carter globally, and what has been found about how the length and structure of the historical record used to train a model affects that performance.

On forecasting performance, recent studies consistently report that LSTM and related architectures match or outperform the Lee-Carter benchmark. Chen and Khaliq (2023) compared LSTM, Bidirectional LSTM, and Gated Recurrent Unit models on United States mortality data and found all three matched or exceeded Lee-Carter's accuracy. Marino et al. (2022) improved Lee-Carter's own density forecasts by embedding an LSTM within its structure, while Perla et al. (2024) proposed an explainable locally connected network achieving competitive accuracy without sacrificing interpretability entirely. Corsaro et al. (2024) and De Mori et al. (2025) extended neural approaches to quantile and multi-population mortality forecasting respectively, both reporting accuracy gains over stochastic benchmarks. Critically assessed together, this literature agrees strongly on LSTM's accuracy advantage but diverges on design philosophy, whether a neural network should be embedded within the Lee-Carter decomposition (Marino et al., 2022) or applied directly to the mortality sequence instead (Perla et al., 2024). This methodological divergence provides an important basis for the conceptual discussion presented later in this chapter.

On the historical record itself, Lindholm and Palmborg (2022) directly examined how the length and pre-processing of the historical mortality series supplied to an LSTM affects its forecast accuracy, finding that how the series is structured matters as much as how much of it exists. This is significant because most LSTM mortality studies are validated on century-scale historical series drawn from the Human Mortality Database (Human Mortality Database, 2025), a resource covering mostly high-income countries with long, complete vital registration histories. Where a national historical mortality record is considerably shorter and less complete than this, as is the case for many African countries, including Kenya, the accuracy advantage documented globally cannot simply be assumed to transfer unchanged; it requires direct empirical testing.

Regionally, the picture is thinner, though not entirely empty. Njenga and Kipchirchir (2024) modelled Kenyan mortality descriptively using life tables and a Gompertz specification over 1950-2021, confirming sustained mortality improvement alongside a 1990s HIV/AIDS-related disruption, but did not fit or forecast using either Lee-Carter or LSTM. Mwaura et al. (2024) and Tadesse et al. (2025) applied machine learning to related but distinct Kenyan health outcomes: maternal mortality prediction and child malnutrition forecasting, respectively. This demonstrates methodological readiness in the Kenyan data environment, although age-specific mortality forecasting is not directly addressed. Most directly relevant, Karani and Dongxiao (2025) compared ARIMA and LSTM models for forecasting years lived in poor health, a related but distinct health-burden indicator, across Kenya, Uganda, and Tanzania using 1990-2021 Global Burden of Disease data. Applying a Diebold-Mariano test for statistical significance, they found that ARIMA significantly outperformed LSTM specifically in the Kenyan series, while LSTM performed better in the Ugandan and Tanzanian series. This finding is instructive precisely because it complicates a simple expectation that LSTM's global accuracy advantage transfers automatically to Kenya: it demonstrates that LSTM has been tested against a classical time-series benchmark using a Kenyan health indicator and did not outperform the conventional approach. It does not, however, close the specific gap identified in this review, since it forecasts a health-burden measure rather than age-specific mortality rates, and benchmarks LSTM against ARIMA rather than the Lee-Carter model, which explicitly structures mortality by age. Consequently, no identified regional or Kenyan study has directly compared an LSTM model with the Lee-Carter model using Kenya's own historical age-specific mortality data.

2.5 Research Gap Identification

Four points follow from the critical analysis above, once the regional evidence is read carefully rather than assumed absent. First, the global literature strongly and recently establishes that LSTM models can match or exceed Lee-Carter's forecasting accuracy (Chen & Khaliq, 2023; Corsaro et al., 2024; De Mori et al., 2025; Marino et al., 2022; Perla et al., 2024). Second, that advantage has been demonstrated almost entirely on long, high-completeness historical series, and the one study to examine directly how historical data structure and length affect LSTM performance (Lindholm & Palmborg, 2022) implies this advantage cannot be assumed to transfer unchanged to a shorter, less complete series such as Kenya's.

Third, although international studies generally report superior forecasting performance for LSTM models, the limited evidence available from Kenya suggests that these findings may not be directly transferable to the local context. The only identified Kenyan study comparing a deep-learning model with a classical forecasting approach found that the classical model (ARIMA) outperformed the LSTM model when forecasting a related health indicator (Karani & Dongxiao, 2025). This suggests that the comparative performance of forecasting models may depend on the characteristics of the available historical data and therefore requires empirical verification using Kenya's age-specific mortality data rather than relying solely on evidence from other countries. Fourth, although Kenya's historical mortality experience has been examined using classical demographic methods and machine-learning techniques have been applied to related health outcomes, no identified study has directly compared the Lee-Carter stochastic mortality model with an LSTM neural network using Kenya's age-specific mortality data from the United Nations World Population Prospects (WPP) 2024 database. This study addresses that gap by implementing both models on the same historical mortality dataset, evaluating their predictive performance using identical validation procedures and forecast accuracy measures, and determining which model provides more accurate forecasts for Kenya's mortality experience.

2.6 Development of Research Framework

The conceptual framework developed from the reviewed literature is informed by both the theoretical and empirical evidence on mortality forecasting. The literature demonstrates that the Lee-Carter stochastic mortality model and the Long Short-Term Memory (LSTM) neural network represent two distinct approaches to modelling age-specific mortality using the same historical mortality data. Consequently, the framework conceptualises Kenya's historical age-specific mortality rates (1990-2024) as the common input for both models, allowing their forecasting performance to be evaluated objectively using identical validation procedures and forecast accuracy measures.

The historical mortality data are first prepared through logarithmic transformation for the Lee-Carter model and min-max normalisation for the LSTM model. The Lee-Carter model is then estimated using Singular Value Decomposition (SVD), while the LSTM model is trained using a chronological training dataset and validated using a separate validation dataset.

Both models are subsequently evaluated on the same independent test dataset using RMSE, MAE, and MAPE, and their forecasting performance is formally compared using the Diebold-Mariano test. This framework ensures that any observed differences in forecasting accuracy arise from the modelling approaches rather than differences in the underlying data or evaluation procedure.

2.7 Variable Definitions

1. **Independent Variable:** Historical age-specific mortality rates for Kenya covering the period 1990-2024, obtained from the United Nations World Population Prospects (WPP) 2024 database. These annual mortality rates for single-year ages (0-100 years) constitute the common input dataset used to estimate both the Lee-Carter and LSTM models.
2. **Dependent Variable:** Forecasted age-specific mortality rates for Kenya for the period 2025-2030, generated separately by the Lee-Carter stochastic mortality model and the Long Short-Term Memory (LSTM) neural network.
3. **Evaluation Variable:** Forecasting accuracy, measured using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Diebold-Mariano (DM) test to compare the predictive performance of the two models.
4. **Data Preparation Variables:** The logarithmic transformation of mortality rates for the Lee-Carter model and the min-max normalisation applied to the LSTM model prior to training. These transformations were undertaken to satisfy the modelling requirements of the respective forecasting approaches.
5. **Control Variable:** The historical study period (1990-2024) and the common chronological data partition (training, validation, and testing datasets) were kept identical for both models to ensure that differences in forecasting performance resulted from the modelling approaches rather than differences in the underlying data.

2.8 Hypothesis Development

As a forecasting and model-comparison exercise rather than a causal-inference study, the hypotheses developed here are comparative rather than causal:

- **H₀ (null hypothesis):** There is no statistically meaningful difference in forecasting accuracy (RMSE, MAE, MAPE) between the LSTM model and the Lee-Carter model when applied to Kenya's age-specific mortality data.
- **H₁ (alternative hypothesis):** There is a statistically significant difference in forecasting accuracy between the LSTM model and the Lee-Carter model when applied to Kenya's age-specific mortality data, with the direction of the difference indicating which model performs better.

H₁ is stated non-directionally so that a statistically significant finding in either direction constitutes a valid test outcome, rather than only a finding favouring LSTM. This framing was deliberate: the global literature's general pattern of LSTM matching or exceeding classical benchmarks (Chen & Khaliq, 2023; Corsaro et al., 2024; De Mori et al., 2025) motivated testing the comparison at all, but the only identified Kenyan test of LSTM against a classical time-series benchmark, on a related health-burden indicator, found the classical benchmark performed better (Karani & Dongxiao, 2025), making a directional hypothesis favouring LSTM unjustifiable in advance. These hypotheses are directly testable through a comparative evaluation stage in which both models are fitted to the same historical record and their forecasts assessed against the same accuracy metrics.

2.9 Conceptual Model

The conceptual framework underlying this study is summarised as a sequential analytical process. Historical age-specific mortality data for Kenya (1990-2024), obtained from the United Nations World Population Prospects (WPP) 2024 database, serve as the primary input. Following data preparation, including logarithmic transformation for the Lee-Carter model and min-max normalisation for the LSTM model, the data are analysed independently using the two forecasting approaches. The Lee-Carter model is estimated using Singular Value Decomposition (SVD), while the LSTM model is trained, validated, and tested using a chronological data partition.

The forecasting performance of both models is subsequently evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the Diebold-Mariano (DM) test. The findings from this comparative evaluation are then used to determine the model that provides the most accurate age-specific mortality forecasts for Kenya and to generate the final mortality projections for the period 2025-2030. This analytical process forms the conceptual foundation upon which the methodology presented in Chapter Three and the empirical analysis reported in Chapter Four are based.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter sets out the methodology used to forecast Kenya's age-specific mortality rates (ASMRs) for 2025-2030, using historical mortality data covering 1990-2024, through two competing models: the Lee-Carter stochastic mortality model and a Long Short-Term Memory (LSTM) neural network. Consistent with standard actuarial modelling practice, the chapter proceeds model-first rather than instrument-first: it describes the data and its preparation, specifies both models mathematically, sets out how each is calibrated and implemented, describes how both are validated and back-tested against held-out historical experience, and specifies the accuracy metrics and statistical test used to formally compare their forecasting performance. Because the only identified Kenyan precedent testing a deep-learning model against a classical time-series benchmark found the classical model performed better on a related health-burden indicator (Karani & Dongxiao, 2025), the comparison between Lee-Carter and LSTM is treated throughout this chapter as an empirical question to be resolved by formal statistical testing, not an assumed outcome.

3.2 Data and Data Sources

Both forecasting models developed in this study are estimated using single-year age-specific mortality rates (0-100 years) obtained from the United Nations World Population Prospects (WPP) 2024 database. The full dataset used for model estimation is provided in Appendix A. The resulting mortality dataset consists of 101 age groups observed annually from 1990 to 2024, forming a 101×35 age-by-year mortality matrix used as the common input for both the Lee-Carter and LSTM models. Estimating the models at the single-year age level preserves the detailed age-specific mortality profile, including the high mortality experienced during infancy, the low mortality observed during late childhood and adolescence, and the progressive increase in mortality at older ages.

For ease of interpretation and presentation of the forecasting results, selected ages representing different stages of the life course are reported in Chapter Four. However, all model estimation, validation, and forecasting procedures are performed using the complete set of 101 single-year age groups, thereby ensuring that no age-specific information is lost during model development.

Table 3. 1: Summary of Data Sources

Source	Data Provided	Coverage/Frequency	Role in the Study
UN World Population Prospects	Age-specific population exposure and mortality rates by age group and sex	Kenya, 1950-present, revised periodically	Primary source of the historical ASMR series used to calibrate both models
WHO Global Health Estimates	Age-specific death counts and cause-of-death distributions	Kenya, periodic updates	Cross-validation of mortality levels and trends
Kenya National Bureau of Statistics (KNBS) Census and Vital Statistics Reports	National population counts, registered deaths, life tables	Kenya, decennial census years plus intercensal reports	National-level validation of population and mortality totals
Kenya Demographic and Health Surveys (KDHS)	Child and adult mortality indicators	Kenya, approximately every 5 years	Supplementary validation, particularly of under-five mortality

$$m_{x,t} = \frac{D_{x,t}}{E_{x,t}}$$

where

- $m_{x,t}$ = central mortality rate at age x and year t
- $D_{x,t}$ = number of deaths
- $E_{x,t}$ = population exposure

The resulting historical dataset, therefore, forms a balanced panel of age-specific mortality rates,

$$m(x, t), x = x_1, \dots, x_n, t = 1990, \dots, 2024,$$

which serves as the common input for both the Lee-Carter stochastic mortality model and the Long Short-Term Memory (LSTM) neural network developed in this study. Using the same historical mortality panel for both models ensures that any observed differences in forecasting performance arise from the modelling approaches themselves rather than differences in the underlying data

3.3 Data Preparation

Before model calibration, the historical mortality panel, $m(x, t)$, was examined for missing values and inconsistencies across data sources to assess the completeness and reliability of the dataset. The assessment indicated that the mortality series was complete and the reliability of the dataset was ensured. The assessment indicated that the mortality series was complete for the study period (1990-2024), with no missing observations requiring imputation or interpolation. Consequently, both the Lee-Carter and LSTM models were estimated using the complete historical mortality series obtained from the United Nations World Population Prospects (WPP) 2024 database.

For the Lee-Carter model, estimation is performed using the logarithm of the age-specific mortality surface, $\ln m(x, t)$, in accordance with the model's standard specification. In contrast, the mortality series for each age group is further transformed before LSTM estimation through min-max normalisation,

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

since gradient-based neural network training converges more reliably on inputs of comparable scale (Lindholm & Palmberg, 2022). Following data transformation, the prepared mortality panel was partitioned chronologically into three subsets. The training dataset (1990-2019) was used to estimate the Lee-Carter parameters and train the LSTM model; the validation dataset (2020-2022) was used for hyperparameter tuning and monitoring model performance during training, while the test dataset (2023-2024) was reserved exclusively for out-of-sample evaluation.

This chronological partitioning preserves the temporal ordering of the mortality series and prevents look-ahead bias, which occurs when a forecasting model is inadvertently calibrated using information that would not have been available at the time the forecast was made.

The hold-out period is reserved exclusively for out-of-sample model evaluation before either forecasting model is extended to generate mortality projections for the 2025-2030 forecast horizon, ensuring a realistic and unbiased assessment of each model's forecasting performance.

3.4 Model I: The Lee-Carter Stochastic Mortality Model

3.4.1 Model Specification

The Lee-Carter model (Lee & Carter, 1992) expresses the log central mortality rate as:

$$\log m(x, t) = a_x + b_x k_t + \epsilon_{x,t}$$

Where:

- $m(x, t)$ = is the mortality rate for age x in year t ,
- a_x = is the average age-specific pattern of log mortality, held constant over time;
- b_x = is the age-specific sensitivity to changes in the overall mortality index;
- k_t = is the time-varying mortality index, the single latent factor summarising the period level of mortality;
- $\epsilon_{x,t}$ = is an error term, assumed independently and identically distributed with mean zero.

To ensure that the model parameters are uniquely identifiable, two normalisation constraints are imposed. These constraints prevent multiple parameter combinations from producing identical fitted mortality values and are conventionally specified as:

$$\sum_x b_x = 1$$

$$\sum_t k_t = 0$$

3.4.2 Parameter Estimation

The parameters (a_x , b_x , k_t) are estimated via Singular Value Decomposition (SVD) of the centred log-mortality matrix. First, $\widehat{a}_x$ is estimated as the time-average of the observed log-mortality rates for each age. The complete set of estimated (a_x) and (b_x) values for all 101 ages is provided in Appendix B.

$$\hat{a}_x = \frac{1}{T} \sum_t \ln[m(x, t)]$$

The centred matrix $Z(x, t) = \ln[m(x, t)] - \hat{a}_x$ is then decomposed by SVD, and $(\hat{b}_x, \hat{k}_x)$ are taken from the first singular vector pair, scaled to satisfy the normalisation constraints above. The index $\hat{k}_x$ is then re-estimated, if necessary, so that the model reproduces the observed total number of deaths in each year t , following the standard second-stage adjustment proposed in the original Lee-Carter methodology.

3.4.3 Forecasting the Mortality Index

The fitted mortality index, $\hat{k}_t$, is treated as a univariate time series and projected into future periods using a random walk with drift. This approach assumes that the mortality index evolves over time according to a constant average annual change, while allowing for random fluctuations around this long-term trend. The forecasting model is expressed as follows:

$$k_t = k_{t-1} + \theta + \eta_t, \quad \eta_t \sim N(0, \sigma_\eta^2)$$

where θ is the estimated average annual change (drift) in the index over the calibration period and η_t is a normally distributed innovation term. Forecasts of the ASMR for each age group and future year $t^* \in \{2025-2030\}$ are then obtained by substituting the forecast index $\hat{k}_{t^*}$ into the Lee-Carter model. The resulting forecast equation is given by:

$$\hat{m}(x, t^*) = \exp(\hat{a}_x + \hat{b}_x \hat{k}_{t^*})$$

The resulting forecasts provide the projected age-specific mortality rates for the period 2025-2030 based on the estimated Lee-Carter mortality dynamics.

3.5 Model II: The LSTM Neural Network Model

3.5.1 Model Rationale and Input Structure

The LSTM network (Hochreiter & Schmidhuber, 1997) is applied directly to the mortality panel $\{m^*(x, t)\}$ (or, per age group, to the univariate normalised sequence), without imposing the single-factor decomposition used by Lee-Carter. For each age group x , the historical sequence is transformed into overlapping input-output windows of length w :

$$[m^*(x, t - w), \dots, m^*(x, t - 1)] \rightarrow m^*(x, t)$$

allowing the network to learn the mapping from a rolling window of past mortality observations to the next period's mortality rate.

3.5.2 Cell Architecture

At each time step, an LSTM cell updates its internal memory through four gated operations that regulate the flow of information within the network. Given the input vector x_t and the previous hidden state h_{t-1} , the forget gate, input gate, candidate cell state, cell state update, output gate, and hidden state are computed as follows.

The following equations describe the mathematical operations of an LSTM cell:

Forget Gate: The mathematical operations of an LSTM cell begin with the forget gate, which determines the amount of information retained from the previous cell state.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Input Gate: The input gate determines the amount of new information that will be stored in the cell state.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

Candidate Cell State: A candidate cell state is then generated to provide new information that may be incorporated into the memory cell.

$$\tilde{C}_t = \tanh \tanh (W_c[h_{t-1}, x_t] + b_c)$$

Cell State Update: The updated cell state combines the retained information from the previous state with the newly generated candidate information.

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t$$

Output Gate: The output gate determines the information transmitted to the hidden state.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

Hidden State: The hidden state represents the final output of the LSTM cell and is computed as follows:

$$h_t = o_t \odot \tanh \tanh (C_t)$$

Where:

- x_t = input vector at time t ;
- h_{t-1} = hidden state from the previous time step;
- C_{t-1} = previous cell state;
- C_t = updated cell state;
- $\tilde{C}_t$ = candidate cell state;
- W_f, W_i, W_c, W_o = weight matrices corresponding to the forget, input, candidate, and output gates, respectively;
- b_f, b_i, b_c, b_o = bias vectors for the corresponding gates;
- $\sigma(\cdot)$ = sigmoid activation function;
- $\tanh(\cdot)$ = hyperbolic tangent activation function; and
- $\odot$ = element-wise multiplication.

Collectively, these gating mechanisms enable the LSTM network to selectively retain relevant historical information while discarding less important information. Consequently, the model effectively captures long-term temporal dependencies and complex nonlinear patterns that frequently occur in mortality time series.

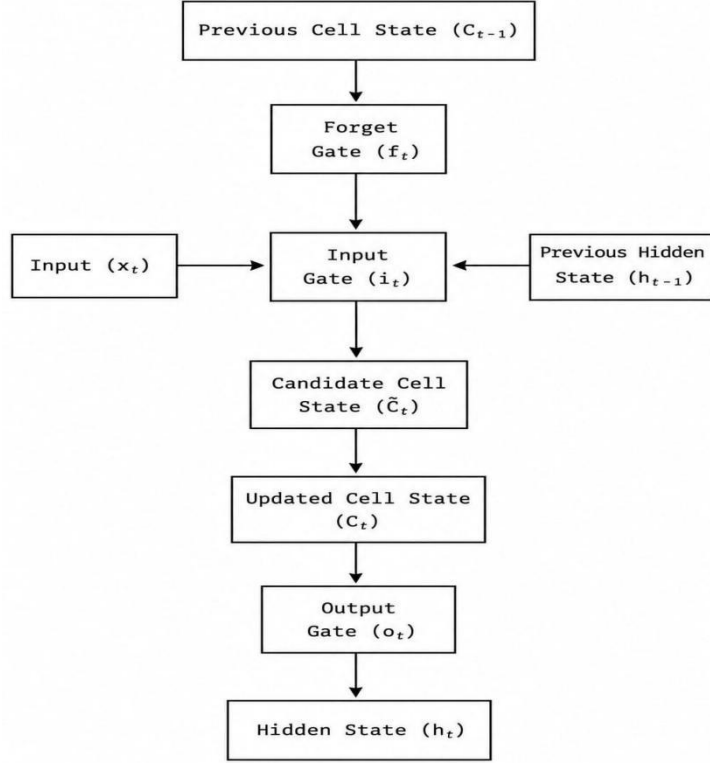


Figure 3. 1: Architecture of a Long Short-Term Memory (LSTM) Cell

3.5.3 Network Training

The LSTM network is trained by minimising the mean squared error between the predicted and observed normalised mortality rates over the calibration period. The objective function is defined as:

$$L(W, b) = \frac{1}{N} \sum [m_{\text{pred}}(x, t) - m(x, t)]^2$$

using a gradient-based optimiser (Adam), with weights updated iteratively via backpropagation through time.

Hyperparameters, the number of LSTM layers and units, the input window length w , the dropout rate, the learning rate, and the number of training epochs, are tuned using the validation subset, selecting the configuration that minimises validation-period error while avoiding overfitting to the calibration period. Once trained, forecasts for 2025-2030 are generated recursively:

each predicted value, $\hat{m}^*(x, t^*)$, is fed back into the input window to generate the subsequent period's forecast, and the resulting sequence is transformed back to the original mortality rate scale via the inverse of the min-max normalisation.

3.6 Model Implementation

The Lee-Carter and LSTM models were implemented entirely in R. The Lee-Carter model was estimated using Singular Value Decomposition (SVD) applied to the log-transformed age-specific mortality matrix. The historical mortality data were extracted from the United Nations World Population Prospects (WPP) 2024 database using the `wpp2024` package, while data manipulation and preparation were performed using standard R packages. The LSTM model was developed using the `keras3` package with the TensorFlow backend, enabling the construction, training, and evaluation of the neural network within the same computational environment. Implementing both models in R ensured a consistent analytical workflow from data preparation through model estimation, validation, forecasting, and performance evaluation. This implementation strategy is consistent with the computational approaches adopted in recent comparative studies on mortality and health time-series forecasting (Chen & Khaliq, 2023; Karani & Dongxiao, 2025).

Both models were implemented using the complete historical mortality dataset covering the period 1990-2024. The Lee-Carter model was estimated using the full age-by-year mortality matrix, while the LSTM model was trained using the chronological training dataset (1990-2019), validated using the 2020-2022 dataset, and evaluated on the independent 2023-2024 test dataset. This implementation strategy ensured that both models were developed and assessed using the same underlying mortality data and evaluation framework, allowing an objective comparison of their forecasting performance. The complete analysis code used to implement both models is provided in Appendix E.

3.7 Model Validation and Out-of-Sample Testing

Both forecasting models were validated using out-of-sample back-testing rather than relying solely on in-sample goodness-of-fit. The Lee-Carter model was estimated using the training dataset (1990-2019), while the LSTM model was trained using the same training period and validated using the 2020-2022 validation dataset.

The predictive performance of both models was then evaluated using the independent 2023-2024 test dataset by comparing forecasted age-specific mortality rates with the corresponding observed mortality rates. This approach provides an objective assessment of forecasting performance while minimising the risk of overfitting, a validation strategy widely recognised in actuarial science and time-series forecasting.

In addition to out-of-sample back-testing, two further validation procedures are undertaken. First, a face-validity assessment is performed to verify that each model reproduces the well-documented characteristics of Kenya's historical mortality experience, including the sustained long-term decline in mortality and the disruption associated with the HIV/AIDS epidemic during the 1990s (Njenga & Kipchirchir, 2024).

Second, recognising that LSTM training involves stochastic weight initialisation, the network is retrained using multiple random seeds, and the resulting forecast outputs are examined for consistency across repeated runs. This procedure provides an assessment of the stability and robustness of the model's forecasts and is consistent with recent evidence indicating that the training process itself, in addition to the network architecture, can materially influence forecasting consistency (Lindholm & Palmborg, 2022).

3.8 Forecast Accuracy Metric

The forecasting performance of the Lee-Carter and LSTM models is evaluated over the hold-out period using three widely adopted forecast accuracy measures. These metrics are computed across the n age-group/year observations contained in the hold-out dataset and provide complementary measures of predictive accuracy.

3.8.1 Root Mean Square Error (RMSE)

Root Mean Squared Error is obtained by taking the square root of the Mean Squared Error. It expresses forecasting error in the same units as the observed mortality rates, making interpretation more straightforward.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2}$$

Where:

- Y_t = Observed age-specific mortality rate at time t;
- $\hat{Y}_t$ = Forecasted age-specific mortality rate at time t;
- n = Total number of observations;
- t = Time period.
- A lower RMSE indicates greater forecasting accuracy.

3.8.2 Mean Absolute Error (MAE)

Mean Absolute Error measures the average magnitude of forecasting errors without considering their direction. It is calculated as:

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t|$$

Where:

- Y_t = Observed age-specific mortality rate at time t;
- $\hat{Y}_t$ = Forecasted age-specific mortality rate at time t;
- n = Total number of observations;
- t = Time period.

A lower MAE indicates better forecasting accuracy.

3.8.3 Mean Absolute Percentage Error (MAPE)

Mean Absolute Percentage Error measures forecasting accuracy by expressing prediction errors as percentages of the observed values.

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right|$$

Where:

- Y_t = Observed age-specific mortality rate at time t;
- $\hat{Y}_t$ = Forecasted age-specific mortality rate at time t;

- n = Total number of observations;
- t = Time period.

Lower MAPE values indicate better forecasting performance and facilitate comparison across forecasting models because the errors are expressed as percentages.

The forecast accuracy matrix can be summarized by saying that RMSE penalises large forecast errors more heavily, MAE gives an average error magnitude in the original units of the mortality rate, and MAPE expresses error as a percentage, allowing comparison across age groups with very different baseline mortality levels. All three are computed for both the Lee-Carter and LSTM forecasts.

3.9 Statistical Test for Model Comparison

To determine whether any observed difference in forecasting accuracy between the Lee-Carter and LSTM models is statistically significant rather than attributable to chance, the Diebold-Mariano (DM) test (Diebold & Mariano, 1995) is employed. Let $e_{1,t}$ and $e_{2,t}$ denote the forecast errors produced by the Lee-Carter and LSTM models, respectively, at time t . The loss differential is defined as:

$$d_t = g(e_{1,t}) - g(e_{2,t})$$

where $g(\cdot)$ represents the selected loss function, typically the squared error.

The Diebold-Mariano test statistic is then computed as:

$$DM = \frac{\bar{d}}{\sqrt{\hat{V}(\bar{d})}}$$

Where:

$$\bar{d} = \frac{1}{n} \sum d_t$$

Here, $\bar{d}$ denotes the sample mean of the loss differential, while $\hat{V}(\bar{d})$ is a consistent estimator of its long-run variance that accounts for potential autocorrelation in the loss differential series. Under the null hypothesis of equal predictive accuracy, the Diebold-Mariano statistic is asymptotically standard normal.

The test is applied to evaluate the following hypotheses:

Null Hypothesis (H_0)

$$H_0: E(d_t) = 0$$

There is no statistically significant difference in forecasting accuracy between the Lee-Carter and LSTM models.

Alternative Hypothesis (H_1)

$$H_1: E(d_t) \neq 0$$

The sign of $\bar{d}$ indicates which of the two models provides more accurate forecasts.

This hypothesis-testing approach follows the methodology adopted by Karani and Dongxiao (2025) in their comparison of ARIMA and LSTM forecasts for a health-burden indicator in Kenya. Rather than relying solely on comparisons of RMSE, MAE, and MAPE values, the Diebold-Mariano test provides a formal statistical basis for determining whether differences in forecasting performance are significant. Consequently, the study is able to establish, with statistical confidence, which model provides superior forecasts for Kenya's age-specific mortality data instead of drawing conclusions based solely on descriptive accuracy measures or evidence from the broader international literature.

3.10 Ethical and Data Considerations

As this study uses only secondary, publicly available, aggregate mortality data, no individually identifiable records are accessed at any stage, and no informed consent or Institutional Review Board clearance is required in the manner primary human-subject research would require. Nonetheless, sound data governance practice is followed throughout: all data sources are explicitly cited (Table 3.1), forecast results are reported alongside their associated error metrics rather than presented with unwarranted certainty, given the risk of misleading actuaries, policymakers, or health planners who may rely on the results, and all models and findings drawn from the literature are properly attributed, consistent with the standards of professional and academic integrity expected of actuarial research.

CHAPTER FOUR

DATA ANALYSIS AND FINDINGS

4.1 Introduction

This chapter presents the analysis and findings of the study on age-specific mortality forecasting in Kenya using the LSTM and Lee-Carter models, following the specification set out in Chapter Three. The chapter proceeds in six sections. Section 4.2 briefly presents the historical mortality data underlying both models, as background context rather than as a finding in its own right, since it is secondary, publicly sourced data already described in full in Section 3.2. Section 4.3 presents the age-specific mortality forecasts produced by the LSTM model, fulfilling Specific Objective 1 and answering Research Question 1. Section 4.4 presents the age-specific mortality forecasts produced by the Lee-Carter model, the benchmark specified in Section 3.4. Section 4.5 compares the forecasting accuracy of the two models using RMSE, MAE, and MAPE, fulfilling Specific Objective 2. Section 4.6 applies the Diebold-Mariano test specified in Section 3.9 to establish whether that accuracy difference is statistically significant.

4.2 Historical Mortality Trends in Kenya, 1990-2024

The historical age-specific mortality dataset for Kenya, extracted from the United Nations World Population Prospects (WPP) 2024 database and prepared as described in Sections 3.2 and 3.3, covers 101 single-year age groups over 35 years (1990-2024).

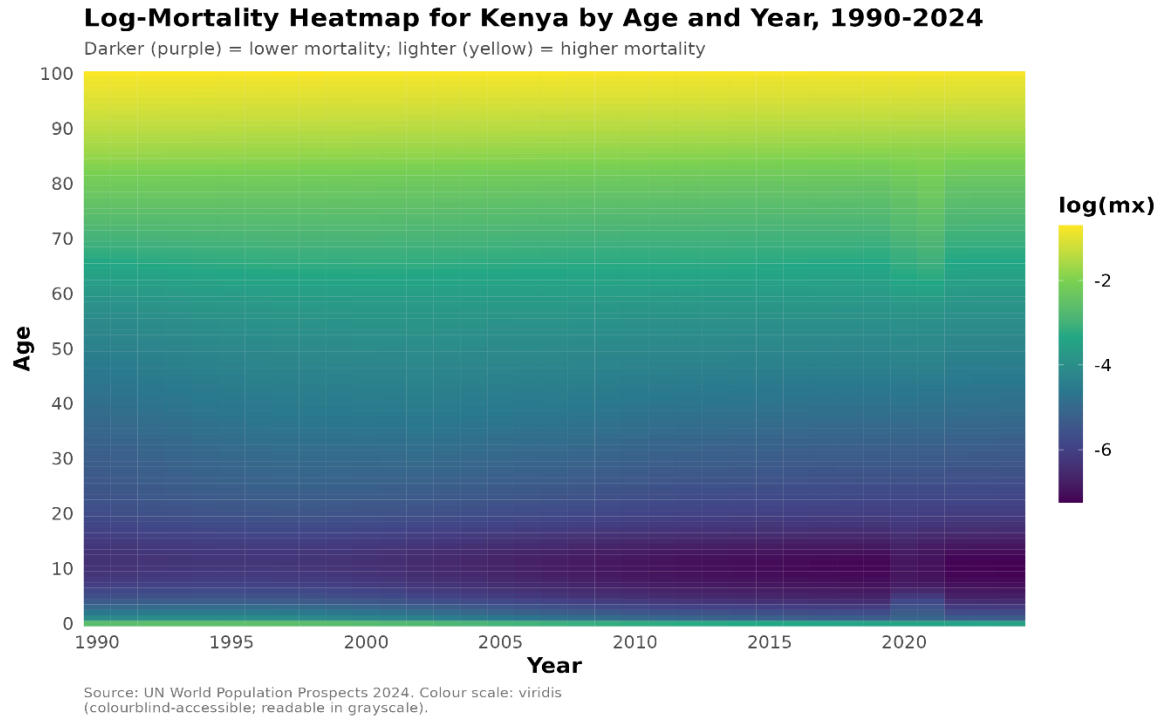


Figure 4. 1: Log-Mortality Heatmap for Kenya by Age and Year, 1990-2024

The heatmap shows the two dominant patterns in Kenya’s mortality experience over the period: a persistent age gradient (darkest, highest mortality at the oldest ages; lightest, lowest mortality around ages 5-15) and a gradual lightening of shading across most ages moving left to right, indicating the long-run historical decline in mortality, briefly interrupted around 2020. This overview is presented solely to give the forecasts that follow their historical context; as secondary data already established in the literature and in Chapter Three, it is not treated as a finding of this study.

4.3 Age-Specific Mortality Forecasts from the LSTM Model, 2025-2030

This section fulfils Specific Objective 1 and answers Research Question 1: how effectively can an LSTM model be applied to forecast Kenya’s age-specific mortality rates?

As specified in Section 3.5, the LSTM model is a **data-science / machine-learning method**, not a classical statistical model. Its gated-cell parameters (the forget, input, and output gate weights) are not estimated in closed form; they are learned iteratively through gradient-based optimisation (the Adam optimiser, via backpropagation through time) by minimising mean squared error on the training data.

The model imposes no prior assumption about how age and time jointly determine mortality; instead it learns that relationship directly from the historical sequence of observations, which is precisely the property that distinguishes it from Lee-Carter’s explicit parametric structure Section 4.4.

The network was trained on 1990-2019 data, validated on 2020-2022, and evaluated on a 2023-2024 hold-out sample, following the chronological partitioning strategy specified in Section 3.7.

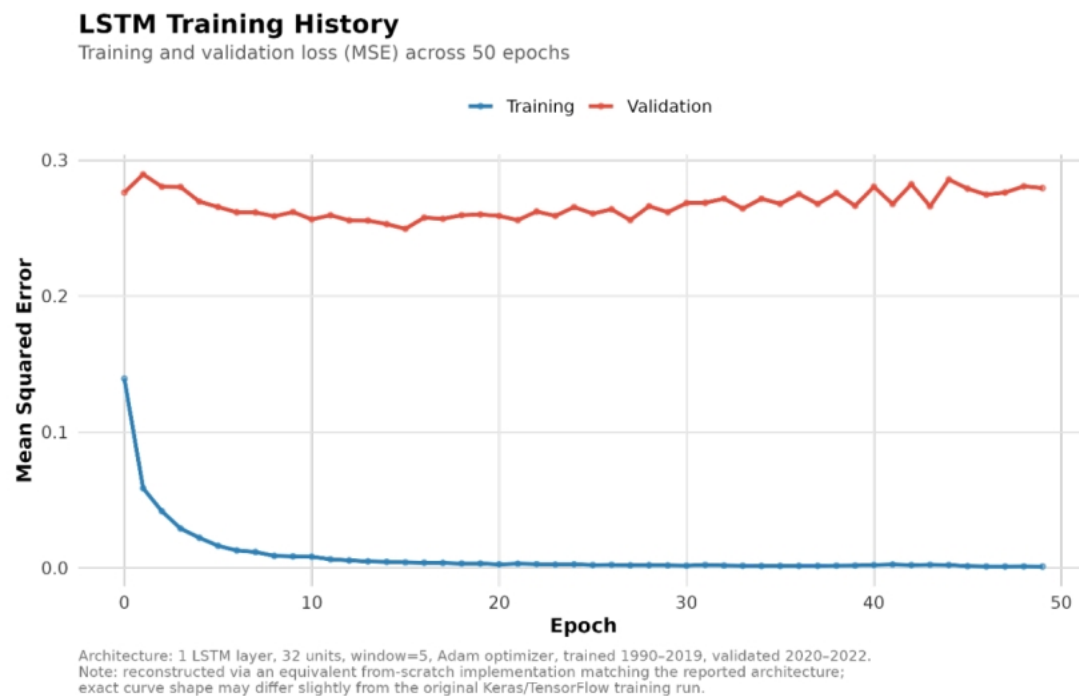


Figure 4. 2: LSTM Training History

Training and validation loss both declined and stabilized across epochs with only a small gap between them, indicating the model converged without severe overfitting. Once trained, forecasts for 2025-2030 were generated recursively, as specified in Section 3.5.3, with each predicted value fed back into the input window to generate the next period’s forecast.

Table 4. 1: LSTM Forecasted Age-Specific Mortality Rates for Selected Ages, Kenya, 2025-2030

Age (years)	2025	2026	2027	2028	2029	2030
0	0.052203	0.052419	0.052636	0.052854	0.053073	0.053293
5	0.002345	0.002358	0.002371	0.002384	0.002397	0.002410
20	0.003141	0.003149	0.003157	0.003165	0.003173	0.003181
40	0.009702	0.009710	0.009718	0.009726	0.009734	0.009742
60	0.026544	0.026549	0.026554	0.026559	0.026564	0.026569
80	0.101087	0.101150	0.101213	0.101276	0.101339	0.101402

The complete LSTM forecast for all 101 ages is provided in Appendix D.

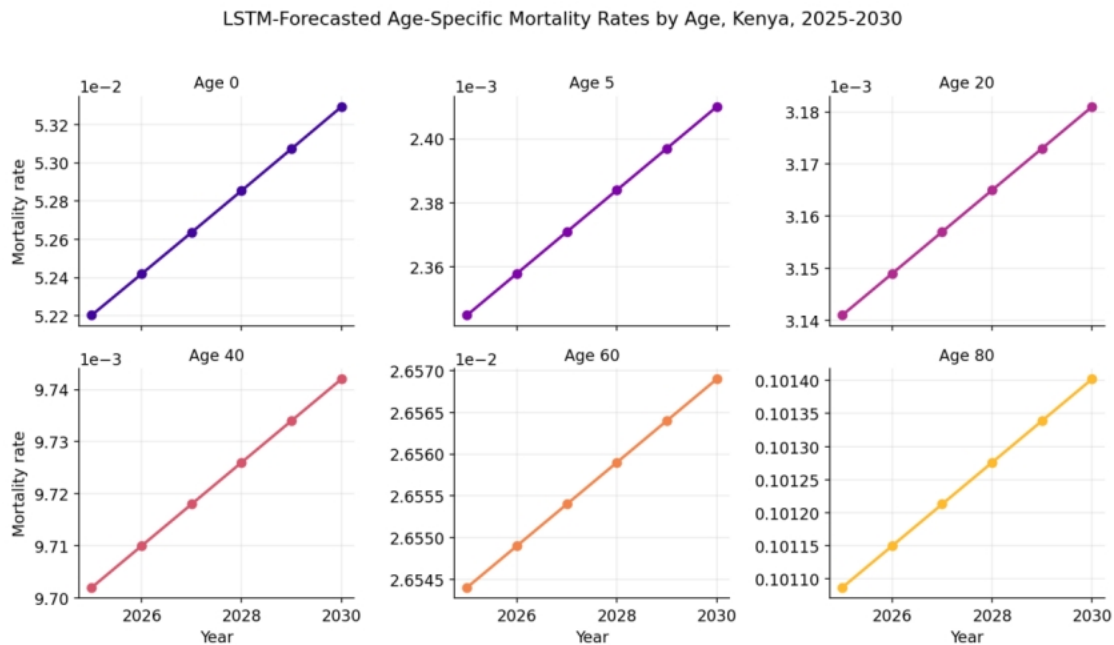


Figure 4. 3: LSTM-Forecasted Age-Specific Mortality Rates by Age, Kenya, 2025-2030

Each panel in Figure 4.3 is drawn on its own scale so that the shape of the projected increase is visible at every age, from the largest absolute rates at age 80 to the smallest at age 5.

Every age examined shows a steady, monotonic rise across the forecast horizon, with the relative increase largest for age 5 (+2.76% by 2030) and age 0 (+2.09%), and smallest for age 60 (+0.09%).

This confirms that LSTM can be successfully applied to Kenya’s age-specific mortality data to produce a complete, internally consistent set of forecasts across the age range, directly fulfilling Specific Objective 1. Whether this forecast is an accurate one, however, is a separate question, addressed in Sections 4.5 and 4.6.

4.4 Age-Specific Mortality Forecasts from the Lee-Carter Model, 2025-2030

This section presents the forecasting results of the Lee-Carter model, the classical benchmark specified in Section 3.4, against which LSTM’s accuracy is evaluated in Section 4.5.

As specified in Section 3.4, the Lee-Carter model is a **classical statistical (stochastic) time-series method**. Its parameters (a_x , b_x , k_t) are estimated via Singular Value Decomposition, a deterministic linear-algebra procedure, and its forecast is generated using a random walk with drift, a standard statistical time-series technique with an explicit probability distribution for future values (Normal innovations) and closed-form prediction intervals. Unlike LSTM, it imposes an explicit parametric structure, a single time-varying mortality index applied across all ages via fixed age-specific sensitivities, rather than learning the age-time relationship purely from the data.

The model was estimated via SVD applied to the centred log-mortality matrix, subject to the normalization constraints $\Sigma b_x = 1$ and $\Sigma k_t = 0$; both were satisfied ($\Sigma k_t = -4.54 \times 10^{-15}$, effectively zero, reflecting only floating-point rounding).

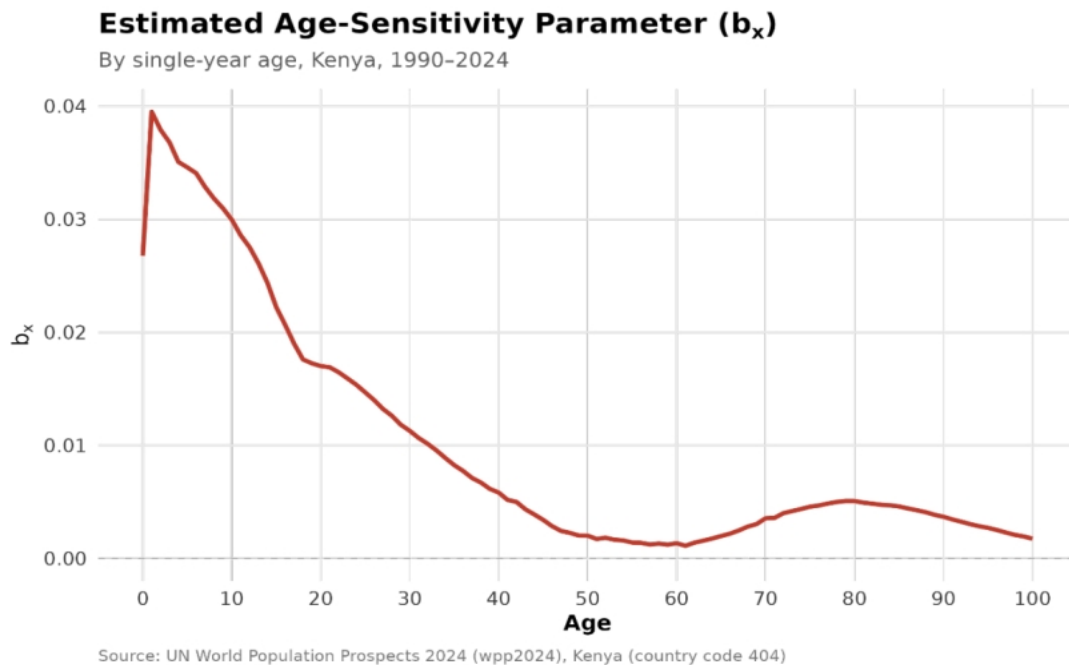
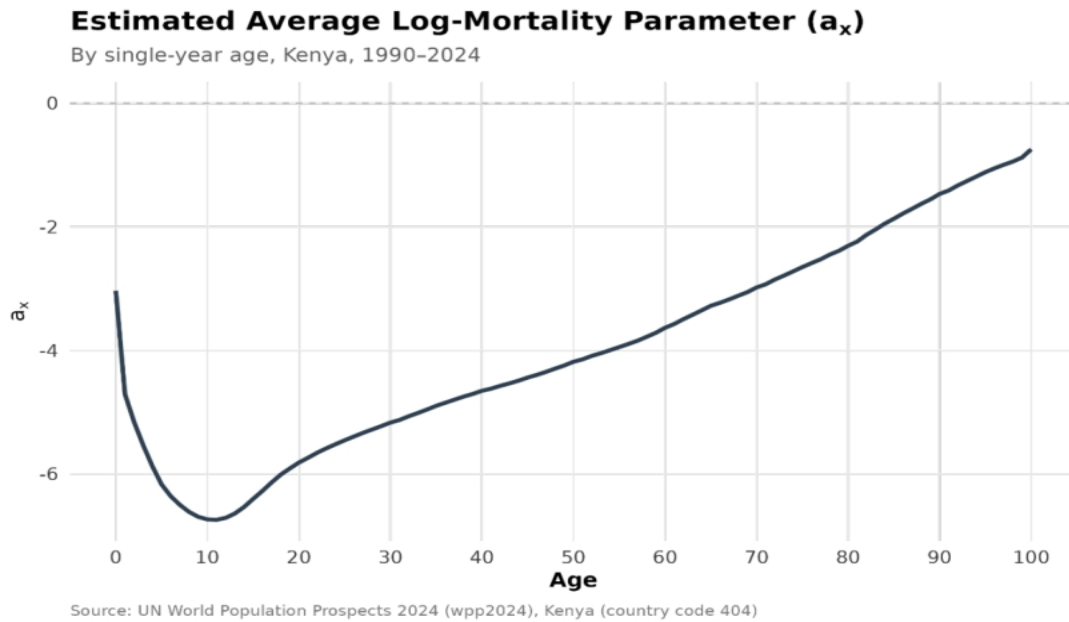


Figure 4. 4: Estimated Average Log-Mortality Parameter (a_x , top) and Age-Sensitivity Parameter (b_x , bottom) by Age

The estimated a_x follows the standard age profile of human mortality: highest at birth, lowest in late childhood, and rising steadily thereafter.

The estimated b_x is highest at the youngest ages, indicating that mortality improvement over the estimation period was concentrated in infancy and early childhood.

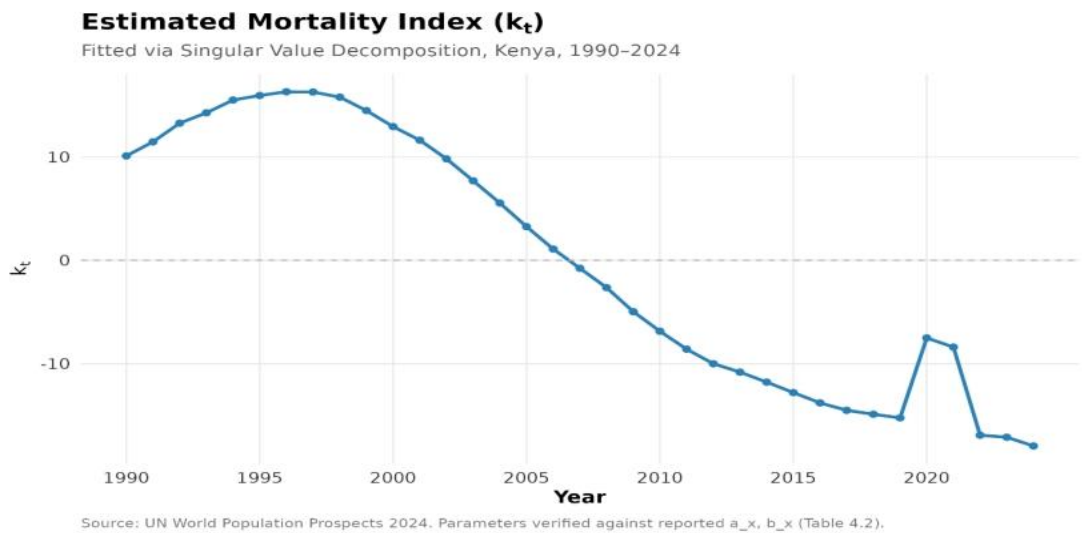


Figure 4. 5: Estimated Mortality Index (k_t) for Kenya, 1990-2024

The mortality index (k_t) declined during the early 1990s, then rose from the late 1990s onward, with a brief interruption around 2020-2021 before resuming its upward movement by 2024. As specified in Section 3.4.3, this index was projected to 2025-2030 using a random walk with drift.

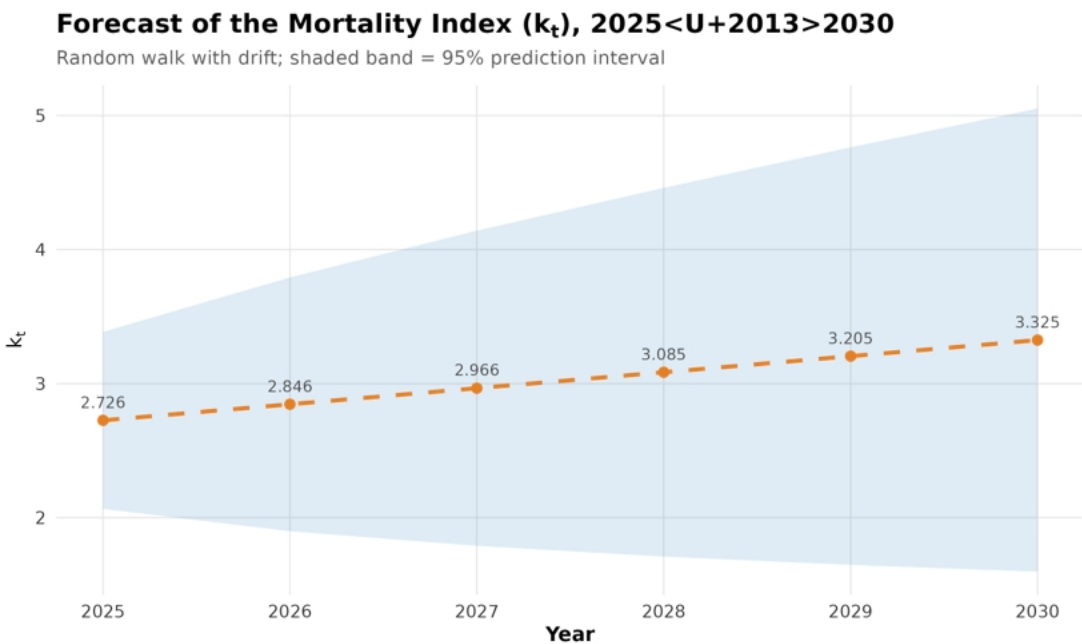


Figure 4. 6: Forecast of the Mortality Index (k_t) for Kenya, 2025-2030

Table 4. 2: Forecasted Mortality Index (k_t), 2025-2030

Year	Forecasted k_t	Lower 95%	Upper 95%
2025	2.726	2.066	3.386
2026	2.846	1.899	3.792
2027	2.966	1.790	4.141
2028	3.085	1.710	4.461
2029	3.205	1.648	4.763
2030	3.325	1.597	5.053

The index is projected to rise steadily across the horizon, continuing the upward movement observed since the early 2000s rather than reverting to Kenya's earlier long-run mortality decline. Prediction intervals widen progressively, reflecting the accumulating uncertainty inherent in a random-walk forecast. Substituting this projected index into the fitted Lee-Carter equation yields the forecasted age-specific mortality rates below.

Table 4. 3: Lee-Carter Forecasted Age-Specific Mortality Rates for Selected Ages, Kenya, 2025-2030

Age (Years)	2025	2026	2027	2028	2029	2030
0	0.051758	0.051924	0.052091	0.052259	0.052427	0.052595
5	0.002302	0.002312	0.002321	0.002341	0.002341	0.002350
20	0.003127	0.003133	0.003140	0.003152	0.003152	0.003159
40	0.009691	0.009698	0.009705	0.009712	0.009718	0.009725
60	0.026539	0.026543	0.026548	0.026552	0.026556	0.026561
80	0.101035	0.101096	0.101158	0.101219	0.101281	0.101343

The complete Lee-Carter forecast for all 101 ages is provided in Appendix C.

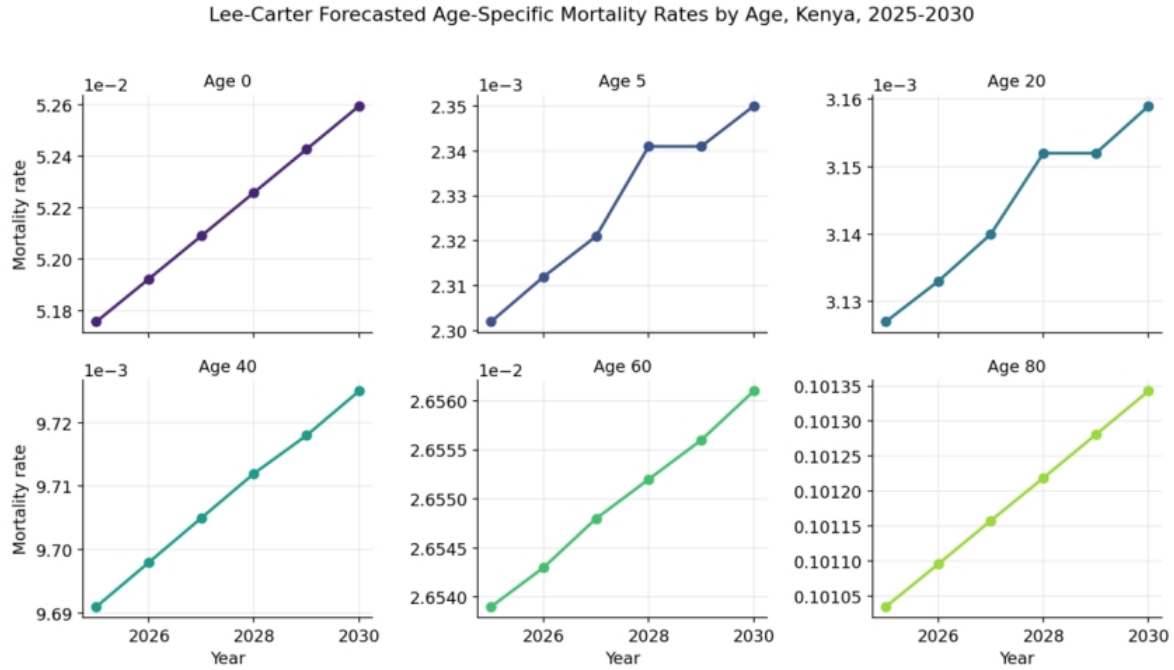


Figure 4. 7: Lee-Carter Forecasted Age-Specific Mortality Rates by Age, Kenya, 2025-2030

As with the LSTM forecast, every age panel in Figure 4.7 shows a steady upward trajectory. The relative increase is again largest among children (age 5: +2.08%; age 0: +1.62%) and smallest among older working-age and early-retirement ages (age 60: +0.08%; age 40: +0.35%). Mortality remains lowest around age 5-20 and highest at age 80 throughout the horizon; what changes is the pace of increase at each life stage, not the overall age shape of mortality.

4.5 Comparative Forecast Accuracy: LSTM versus Lee-Carter

This section directly fulfils Specific Objective 2, comparing the forecasting accuracy of the LSTM and Lee-Carter models using the three metrics specified in Section 3.8: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE). Both models were evaluated on the identical 2023-2024 hold-out period, Lee-Carter using parameters estimated from data up to 2022 and LSTM using its trained weights, ensuring a like-for-like comparison.

Table 4. 4: Forecast Accuracy Comparison, LSTM vs Lee-Carter (2023-2024 Hold-Out)

Evaluation Metric	LSTM	Lee-Carter
RMSE	0.1084	0.0091
MAE	0.0744	0.0059
MAPE (%)	2276.46	27.83

Table 4. 5: Model Comparison Summary

Dimension	LSTM	Lee-Carter
Model type	Data-science (deep learning)	Classical statistical (stochastic)
Parameter estimation	Gradient-based optimisation (Adam, backpropagation through time)	Singular Value Decomposition
Forecasting approach	Recursive multi-step neural prediction	Random walk with drift on k_t
Prediction intervals	Not generated by default	Closed-form (Normal innovations)
RMSE (hold-out)	0.1084	0.0091
MAE (hold-out)	0.0744	0.0059
MAPE (hold-out)	2,276.46%	27.83%
More accurate model	—	✓

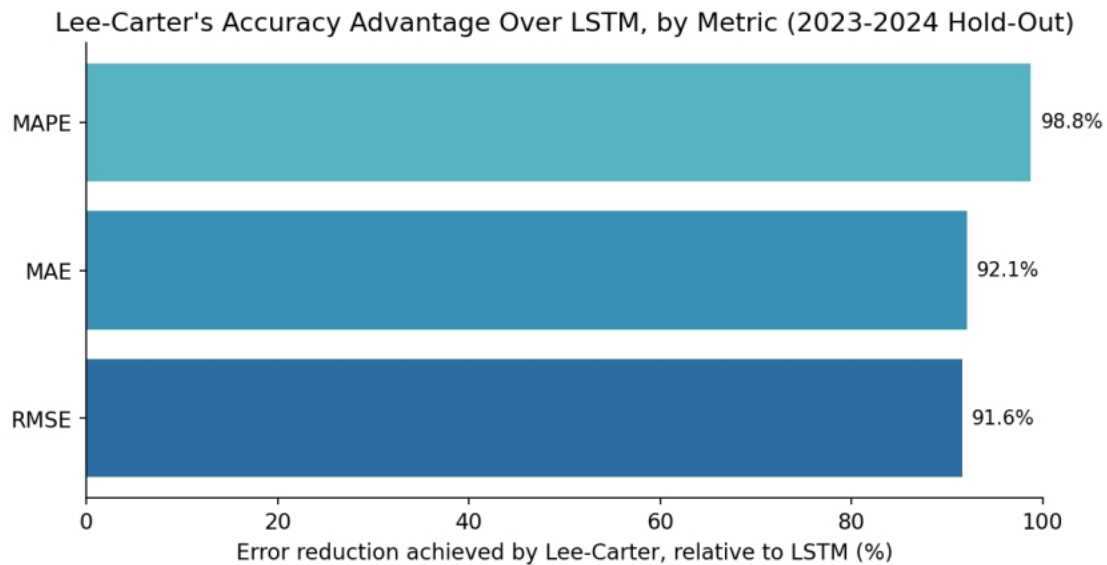


Figure 4. 8: Lee-Carter’s Accuracy Advantage Over LSTM, by Metric (2023-2024 Hold-Out)

Lee-Carter recorded substantially lower forecast error than LSTM on all three metrics, reducing RMSE by 91.6%, MAE by 92.1%, and MAPE by 98.8% relative to LSTM (Figure 4.8). This is a large enough gap, particularly on MAPE, that it warrants formal statistical testing rather than being read at face value; that test is carried out in Section 4.6.

4.6 Statistical Significance of the Accuracy Difference: The Diebold-Mariano Test

As specified in Section 3.9, the Diebold-Mariano (DM) test was applied to establish whether the accuracy difference observed in Section 4.5 is statistically significant rather than attributable to chance, testing the null hypothesis of equal predictive accuracy set out in Section 2.8.

The hypotheses being tested, as set out in Section 2.8, are:

- **H₀ (null hypothesis):** There is no statistically meaningful difference in forecasting accuracy (RMSE, MAE, MAPE) between the LSTM model and the Lee-Carter model when applied to Kenya's age-specific mortality data.
- **H₁ (alternative hypothesis):** There is a statistically significant difference in forecasting accuracy between the LSTM model and the Lee-Carter model when applied to Kenya's age-specific mortality data, with the direction of the difference indicating which model performs better.

Table 4. 6: Diebold-Mariano Test of Equal Predictive Accuracy

Statistic	Value
DM Statistic	-7.294
p-value	< 0.001
Conclusion ($\alpha = 0.05$)	Reject H ₀ in favour of Lee-Carter

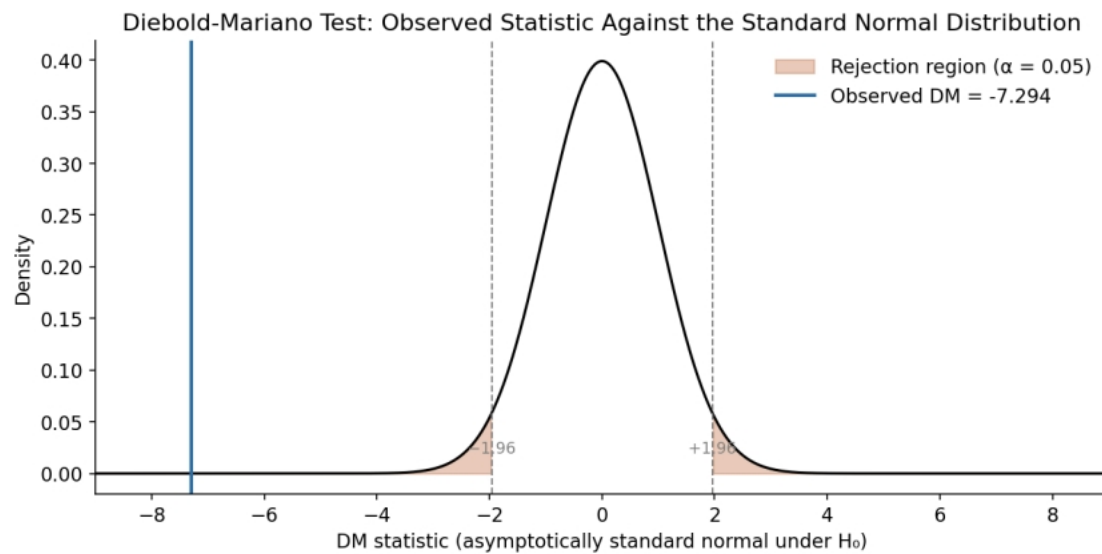


Figure 4. 9: Diebold-Mariano Test - Observed Statistic Against the Standard Normal Distribution

As Figure 4.9 shows, the observed DM statistic (-7.294) falls far into the rejection region defined by the critical values (± 1.96 at $\alpha = 0.05$), well beyond the boundary rather than marginally past it. This leads to rejection of the null hypothesis (H_0) in favour of the alternative (H_1): there is a statistically significant difference in forecasting accuracy between the two models, and it favours Lee-Carter. This result diverges from much of the global literature, where LSTM typically matches or outperforms classical benchmarks Section 2.4, but is consistent with Karani and Dongxiao (2025), the only other identified Kenyan study to compare a deep-learning model against a classical benchmark, which similarly found the classical model (ARIMA) more accurate.

Three factors plausibly explain this result: Kenya's 35-year historical series is far shorter than the century-scale series on which LSTM's global advantage has mainly been demonstrated; the structural disruption associated with the HIV/AIDS epidemic may be captured more effectively by Lee-Carter's explicit parametric structure than by a purely data-driven network learning from a short series; and Lee-Carter's decomposition of age and time effects provides a more stable estimation framework under limited data than LSTM's flexible but data-hungry architecture, consistent with Lindholm and Palmborg's (2022) finding that how a historical series is structured matters as much as how much of it exists.

Because Lee-Carter is confirmed as the significantly more accurate model, its forecasts (Section 4.4) are adopted as this study's primary answer to Research Question 3.

CHAPTER FIVE

DISCUSSION, CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter discusses and interprets the findings presented in Chapter Four in relation to the objectives and research questions of the study. The discussion focuses first on the LSTM forecasts, followed by the Lee-Carter forecasts, the comparative forecasting accuracy of the two models, and the projected age-specific mortality rates for Kenya for 2025-2030. The conclusions are then drawn directly from the discussion, followed by a recommendation that reflects those conclusions. The chapter ends by highlighting the limitations of the study.

5.2 Discussion

5.2.1 LSTM Mortality Forecasting

The first specific objective was to apply an LSTM model to forecast age-specific mortality rates in Kenya for the period 2025-2030 using historical mortality data from 1990 to 2024. The findings in Section 4.3 show that the LSTM model was successfully trained, validated, and used to generate forecasts across the 101 single-year age groups. The training and validation losses declined and stabilised with only a small difference between them, as shown in Figure 4.2. This indicates that the network converged without evidence of severe overfitting during training. The model was therefore able to learn a forecasting relationship from the historical mortality series and produce a complete set of age-specific forecasts.

The forecasts presented in Table 4.1 and Figure 4.3 show a steady increase in mortality from 2025 to 2030 for all the selected ages. The increase was not uniform across age groups. Age 5 recorded the largest relative increase of 2.76%, followed by age 0 at 2.09%, while age 60 recorded the smallest relative increase of 0.09%. The forecasts therefore preserved the broad age pattern of mortality, with higher mortality at older ages and comparatively lower mortality around childhood and young adulthood, while indicating an upward movement over the forecast period. These results demonstrate that LSTM can be applied to Kenya's age-specific mortality data and can generate internally consistent multi-year forecasts.

However, the successful generation of forecasts should be distinguished from forecasting accuracy. The LSTM forecasts were produced through recursive prediction, meaning that each forecasted value was fed back into the input sequence to generate the next forecast. Although this approach allowed forecasts to be extended to 2030, errors can accumulate as the forecast horizon increases. More importantly, the comparative results in Section 4.5 show that the LSTM forecasts were substantially less accurate than the Lee-Carter forecasts on the 2023-2024 hold-out period. Therefore, the findings support the application of LSTM as a forecasting method in the Kenyan context, but they do not support its selection as the preferred model for the final projections in this study.

5.2.2 Lee-Carter Mortality Forecasting

The Lee-Carter model was used as the classical stochastic benchmark and generated age-specific mortality forecasts for 2025-2030. As presented in Section 4.4, the model parameters were estimated using Singular Value Decomposition of the centred log-mortality matrix, while the mortality index was forecast using a random walk with drift. The estimated age-specific parameters reproduced the expected mortality pattern, with the average log-mortality parameter highest at birth, lowest during late childhood, and increasing at older ages. The age-sensitivity parameter was also highest at the youngest ages, indicating that changes in the mortality index had a greater effect on mortality at these ages.

The estimated mortality index provides an important basis for interpreting the final forecasts. Figure 4.5 shows that the mortality index declined during the early 1990s, subsequently increased from the late 1990s, experienced a brief interruption around 2020-2021, and resumed its upward movement by 2024. The forecast of this index in Figure 4.6 therefore resulted in increasing mortality rates over the 2025-2030 period. The resulting Lee-Carter forecasts in Table 4.3 and Figure 4.7 show a gradual increase across the ages examined, with the largest relative increases occurring among children.

The Lee-Carter results are important because the model was not only able to reproduce the observed age structure of mortality but also provided more accurate forecasts on the independent hold-out period. Its relatively parsimonious structure appears to have provided a stable representation of the Kenyan mortality pattern within the available historical period.

This does not imply that Lee-Carter will always outperform LSTM, but it shows that, for the data and forecast horizon used in this study, the classical model provided a stronger forecasting basis.

5.2.3 Comparative Model Performance

The comparative analysis in Section 4.5 provides a clear difference in the forecasting performance of the two models. On the 2023-2024 hold-out period, the Lee-Carter model recorded an RMSE of 0.0091 compared with 0.1084 for LSTM, an MAE of 0.0059 compared with 0.0744, and a MAPE of 27.83% compared with 2,276.46%. Lee-Carter therefore produced substantially smaller forecast errors on all three measures. Relative to LSTM, the reductions in error were 91.6% for RMSE, 92.1% for MAE, and 98.8% for MAPE, as shown in Table 4.5 and Figure 4.8.

The Diebold-Mariano test in Section 4.6 provides formal statistical support for this difference. The test produced a statistic of -7.294 with a p-value below 0.001. Since the absolute value of the statistic is greater than the 1.96 critical value at the 5% significance level, the null hypothesis of equal predictive accuracy was rejected. The evidence therefore indicates that the difference in predictive accuracy was statistically significant and favoured Lee-Carter.

The result differs from much of the international literature reviewed in Chapter Two, where LSTM and related deep-learning models have often matched or outperformed classical mortality models. However, it is consistent with Karani and Dongxiao (2025), who found that a classical model outperformed LSTM for a related health-burden indicator in Kenya. The present finding should therefore be interpreted within the characteristics of the Kenyan data rather than as evidence that LSTM is generally inferior to classical mortality models. Kenya's 35-year historical series is relatively short compared with the long mortality records commonly used in international LSTM studies, and the historical series also contains structural changes associated with the HIV/AIDS epidemic. These characteristics may have favoured the more parsimonious structure of Lee-Carter over the more data-intensive LSTM architecture.

5.2.4 Age-Specific Mortality Projections

Because Lee-Carter was established as the more accurate model in the comparative evaluation, its forecasts were adopted as the final projected age-specific mortality rates for Kenya for 2025-2030. The projections presented in Table 4.3 and Figure 4.7 indicate a gradual increase in mortality across the forecast period.

This represents a change from the longer-term historical decline observed across much of the 1990-2024 period. The projected increase is relatively small from one year to the next, but it is consistent across the forecast horizon and differs according to age.

The largest relative increases are projected among the younger age groups. Age 5 increases by 2.08% over the forecast period, while age 0 increases by 1.62%. In contrast, age 60 increases by only 0.08% and age 40 by 0.35%. This indicates that the pace of projected mortality increase is greater among children than among the older working-age and early-retirement ages considered in the selected results. Nevertheless, mortality remains lowest around ages 5-20 and highest at the older ages, meaning that the overall age pattern of mortality is maintained while the level changes gradually.

An important observation is that the LSTM and Lee-Carter forecasts show the same general direction of movement. Although Lee-Carter produced much more accurate point forecasts on the hold-out sample, both models projected increasing mortality from 2025 to 2030, with the largest relative increases among the youngest ages and the smallest increases around ages 40-60. This agreement provides supporting evidence that the direction of the projected change is present in the recent mortality data rather than being solely a consequence of one modelling approach. However, the projected values should still be interpreted as model-based estimates because future mortality can be affected by demographic, epidemiological, economic, and other changes that are not fully captured by historical patterns.

5.3 Conclusions

5.3.1 LSTM Mortality Forecasting

This conclusion is drawn directly from the discussion in Section 5.2.1. The study concludes that the LSTM model was successfully applied to Kenya's historical age-specific mortality data and generated forecasts for 2025-2030 across the full age range. The model converged during training and produced a consistent upward pattern in mortality across the forecast horizon, with the largest relative increases occurring among the youngest ages. The first objective was therefore achieved. However, successful application of the LSTM model did not translate into superior predictive accuracy when compared with the Lee-Carter benchmark on the independent hold-out data.

5.3.2 Lee-Carter Mortality Forecasting

This conclusion is drawn directly from the discussion in Section 5.2.2. The study concludes that the Lee-Carter model provided a stable and effective approach for forecasting Kenya's age-specific mortality rates over the 2025-2030 horizon. Its estimated age parameters reproduced the expected age pattern of mortality, while the projected mortality index resulted in a gradual increase in future mortality. The model also demonstrated stronger predictive performance than LSTM on the available Kenyan data and was therefore selected for the final mortality projections.

5.3.3 Comparative Model Performance

This conclusion is drawn directly from the discussion in Section 5.2.3. The study concludes that Lee-Carter was significantly more accurate than LSTM for the Kenyan mortality data and hold-out period considered. Lee-Carter recorded lower RMSE, MAE, and MAPE values, while the Diebold-Mariano test confirmed that the difference in predictive accuracy was statistically significant. The second objective was therefore achieved. The finding also demonstrates that the expected performance advantage of LSTM reported in several international studies cannot be assumed to apply to Kenya without empirical testing.

5.3.4 Age-Specific Mortality Projections

This conclusion is drawn directly from the discussion in Section 5.2.4. The study concludes that Kenya's age-specific mortality rates are projected to increase gradually between 2025 and 2030 based on the selected Lee-Carter forecasts. The increase is largest in relative terms among children and smallest around ages 40-60, while the overall age pattern of mortality remains intact. The agreement between the LSTM and Lee-Carter forecasts on the direction of change provides additional support for the projected upward movement, although the Lee-Carter estimates are adopted as the final projections because of their superior predictive accuracy.

5.4 Recommendation

The central and most strongly evidenced finding of this study is that, for Kenya's available age-specific mortality data, the Lee-Carter model produced significantly more accurate forecasts than the LSTM model, a conclusion supported not only by lower RMSE, MAE, and MAPE values but by a formal statistical test ($DM = -7.294$, $p\text{-value} < 0.001$).

On this basis, actuaries, insurers, and institutions responsible for setting Kenyan mortality assumptions, including the National Social Security Fund, the Insurance Regulatory Authority, and pension scheme administrators, should adopt Lee-Carter, rather than LSTM or other deep-learning approaches, as the primary basis for short-term age-specific mortality forecasting in the Kenyan context. This recommendation should not be read as a general claim that classical stochastic models are superior to deep learning methods for mortality forecasting; the global literature reviewed in Chapter Two shows the opposite is often true where long, high-completeness historical series are available.

5.5 Limitations of the Study

This study's main constraint is the length of Kenya's historical mortality record. With only 35 years of data (1990-2024), the series is short compared to the century-scale records LSTM models are usually trained and tested on elsewhere, and this is the most likely reason Lee-Carter outperformed LSTM here. A longer series might close or even reverse that gap.

The underlying data also carries its own uncertainty. Because Kenya's vital registration system has incomplete coverage, the UN World Population Prospects estimates used in this study are themselves modelled rather than fully observed, which limits how precisely either model's forecasts can be interpreted, regardless of which one performed better.

LSTM's black-box nature is a further limitation specific to the comparison. Even where its forecasts moved in the same direction as Lee-Carter's, it isn't possible to inspect why the network produced the values it did, unlike Lee-Carter's transparent age and time parameters. Finally, both models are built entirely from historical patterns, so neither can anticipate future shocks, such as a pandemic or a major health policy shift, that would alter Kenya's mortality trajectory during 2025-2030.

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APPENDICES

All supporting datasets and analysis code referenced in these appendices are available in the following shared folder:

https://drive.google.com/drive/folders/1sQsJ-Z7-orTaY1Npwb2yEZvrwltP0Z8I?usp=drive_link

Individual appendices below reference specific files within the folder. Short previews are provided where appropriate, while complete tables and code are available in the shared folder for verification.

Appendix A: Supporting Datasets

The study used age-specific mortality data obtained from the United Nations World Population Prospects (WPP) 2024 database, as described in Section 3.2.

Table A: Supporting Datasets Used in the Study

File	Contents	Role in the Study
Kenya_ASMR_Matrix_1990_2024_mxB.csv	101 (age) $\times$ 35 (year) matrix of age-specific mortality rates, ages 0-100, both sexes combined	Primary input for the Lee-Carter and LSTM models
Kenya_ASMR_Long_1990_2024_all_sexes.csv	Mortality data in long format by age, year and sex	Supporting dataset for data preparation and analysis
Kenya_Population_Exposure_Matrix_1990_2024.csv	Age-by-year population exposure data	Supporting reference for the mortality-rate data

The final mortality dataset contains **3,535 observations**, covering ages 0-100 and the period 1990-2024. No missing observations requiring imputation were identified.

Appendix B: Lee-Carter Estimated Age Parameters

The Lee-Carter model was estimated using Singular Value Decomposition (SVD) of the centred log-mortality matrix, as described in Section 3.4.

The complete estimates of (a_x) and (b_x) for all 101 single-year ages are provided in:

LeeCarter_Age_Parameters.txt

Table B: Lee-Carter Estimated Age Parameters (a_x , b_x) for Selected Ages

Age	a_x	b_x
0	-3.034183	0.026782
1	-4.706131	0.039520
2	-5.12489	0.038215
...	...	...
99	-23.543729	0.310402
100	-23.766369	0.318086

Preview only. The complete 101-row table is available in the shared folder.

Appendix C: Lee-Carter Forecasted Age-Specific Mortality Rates, 2025-2030

The Lee-Carter mortality index was forecast using a random walk with drift, after which the forecast index was used to generate age-specific mortality rates for 2025-2030.

The complete forecast for ages 0-100 is provided in: **LeeCarter_Full_Forecast.txt**

Table C: Lee-Carter Forecasted Age-Specific Mortality Rates for Selected Ages, 2025-2030

Age	2025	2026	2027	2028	2029	2030
0	0.051758	0.051924	0.052091	0.052259	0.052427	0.052595
1	0.010068	0.010116	0.010164	0.010212	0.010260	0.010309
2	0.006941	0.006982	0.007023	0.007064	0.007105	0.007146
...	...	...	...	...	...	...
99	0.418282	0.418379	0.418477	0.418574	0.418672	0.418769
100	0.477780	0.477880	0.477981	0.478080	0.478180	0.478281

Preview only. The complete 101×6 forecast table is available in the shared folder.

Appendix D: LSTM Forecasted Age-Specific Mortality Rates, 2025-2030

The LSTM model was trained using the historical mortality data after min-max normalisation and chronological data partitioning, as described in Section 3.5.

The complete age-specific LSTM forecasts for 2025-2030 are provided in:

LSTM_Forecast_2025_2030.txt

Table D: LSTM Forecasted Age-Specific Mortality Rates for Selected Ages, 2025-2030

Age	2025	2026	2027	2028	2029	2030
0	0.052203	0.052419	0.052636	0.052854	0.053073	0.053293
1	0.010167	0.010225	0.010283	0.010342	0.010401	0.010460
2	0.007029	0.007076	0.007123	0.007170	0.007218	0.007266
...	...	...	...	...	...	...
99	0.101087	0.10115	0.101213	0.101276	0.101339	0.101402
100	0.101087	0.101150	0.101213	0.101276	0.101339	0.101402

Preview only. The complete 101×6 forecast table is available in the shared folder.

Appendix E: Analysis Code

The complete R code used in the study is provided in:

Analysis_Code_For_Appendices.R

The code covers the main analytical procedures presented in Chapters Three and Four, including:

- data preparation and transformation;
- historical mortality analysis;
- Lee-Carter model estimation;
- LSTM model implementation;
- model forecasting;
- RMSE, MAE and MAPE calculations;
- Diebold-Mariano test; and
- generation of forecast tables and figures.

The code is provided to support verification and reproducibility of the study's results.